

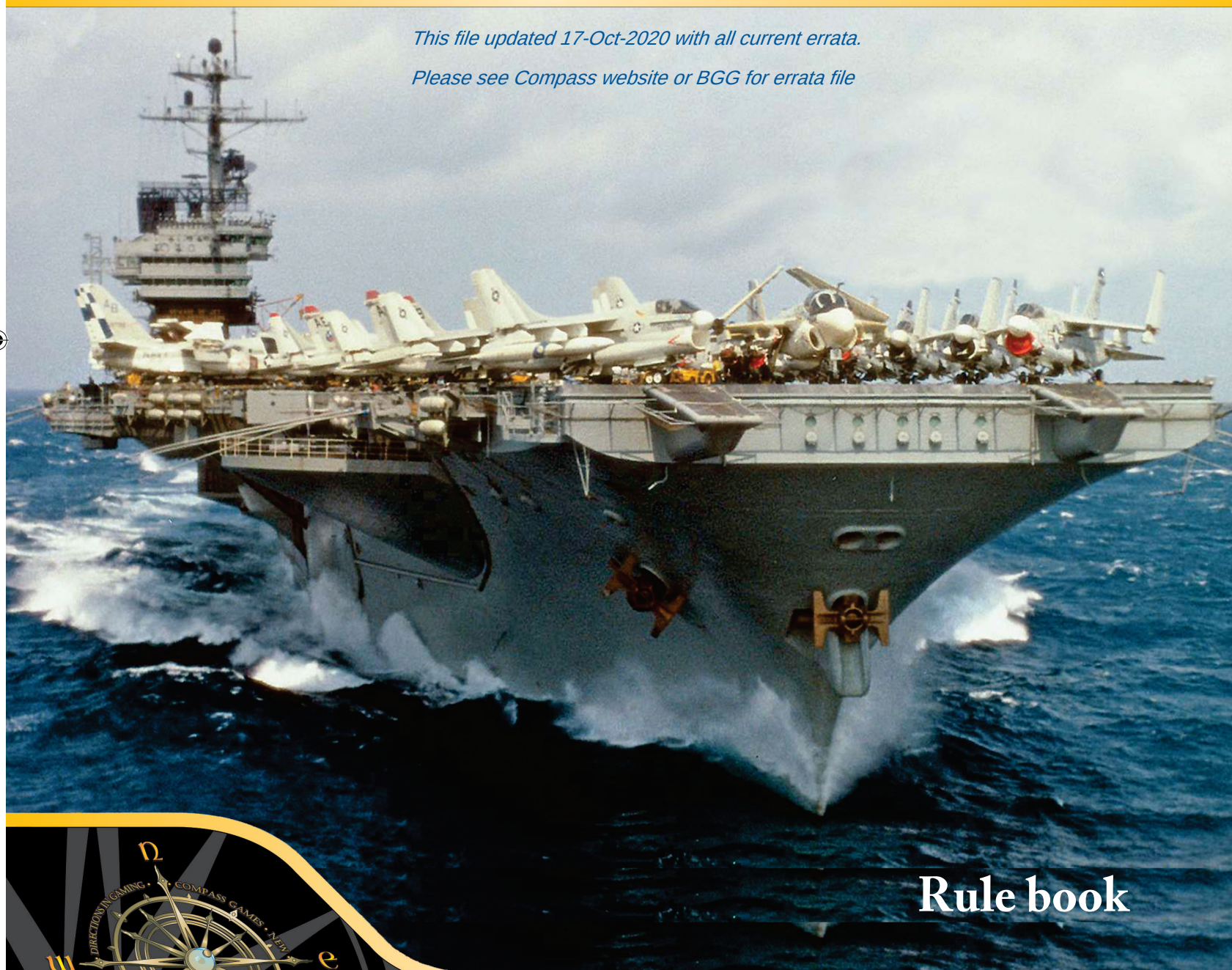


BLUE WATER NAVY

The War at Sea

This file updated 17-Oct-2020 with all current errata.

Please see Compass website or BGG for errata file



Rule book

Compass Games
New Directions in Gaming

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GLOSSARY

Action	– The expenditure of one or more Operations Point(s) for a unit to perform a task.
Combined Facility	– A port and airfield with a single damage box printed on the map.
Amphib	– Amphibious unit – a group of 20+ ships for landing ground troops. NATO amphib transports troops to NATO ports for landing where as Soviet amphib transports troops to landing sites printed on the map.
NATO	– North Atlantic Treaty Organization.
Bad Weather	– Sea state 6 and above (defined as ‘very rough’).
Natural Roll	– The value on a die before any DRMs are applied.
Damage Box	– An information box printed next to each facility showing the number of hits required to cause damage.
Operation Points (Ops) Also ‘OP’	– Points derived from cards used to perform actions.
DRM	– Die Roll Modifier.
Sea Zone	– Each sea area on the map.
FSP	– First Strike Point.
Hydrophone barrier	– Listening devices deployed on the ocean floor for detecting submarines.
Soviet	– The Union of Soviet Socialist Republics (USSR).
Invasion Tracks	– The tracks representing fighting in Norway and Denmark, and the focus of Amphibious Operations in the game.
Turn	– A turn includes the play and resolution of two cards per player with some additional starting and ending steps. Each card represents one day.
Land Area	– An area of land adjacent to a Sea Zone.
War Tracks	– The tracks representing the fighting in Germany, Italy, Turkey, Belgium, Netherlands and France.
Maritime Patrol (MP)	– Aircraft designed to patrol and detect Task Forces and submarines.
Weather Zone	– There are 5 Weather Zones on the map to regulate the appearance of Bad Weather.

MILITARY ACRONYMS

AAA	Anti-Air Artillery	Guns & SAMs designed to shoot down aircraft.
AEW	Airborne Early Warning	Airborne RADAR platforms for detecting aircraft and ships.
ASM	Anti-Ship Missile	A missile designed to sink surface ships.
ASW	Anti-Submarine Warfare	The detection, classification and destruction of enemy submarines.
ASuW	Anti-Surface Warfare	The detection and destruction of enemy shipping
CAP	Combat Air Patrol	Fighter aircraft flying a patrol to detect and destroy enemy strike aircraft.
CAPTOR	Encapsulated Torpedo	An American mine with built-in SONAR system, automatically detects and attacks enemy submarines, releasing a lightweight torpedo.
ECM	Electronic Countermeasures	Electronic devices designed to reduce enemy RADAR effectiveness.
Hydrophones		An underwater microphone to listen for noise – see SONAR. Used by air units to deploy detection ‘nets’ into the sea.
RADAR	Radio Detection And Ranging	A method for detecting distant objects using radio waves – in naval combat, it is used to detect surface ships and aircraft.
RORSAT	Soviet Satellite	RADAR Ocean Reconnaissance Satellite. A nuclear-powered satellite with a powerful RADAR transmitter designed to detect NATO fleet units at sea. Often paired with an EOSAT (Electro Optical Satellite) used for detecting emitted signals such as communications and RADAR.

SAM	Surface to Air Missile	A missile fired from the surface and designed to destroy aircraft and ASMs.
SEAD	Suppression of Enemy Air Defenses	The neutralization of air defense networks either via their destruction or suppression such that they do not interfere with air strikes.
SONAR	Sound Navigation And Ranging	A method for detecting distant objects using sound waves – submarines and aircraft use sound to detect other submarines
SOSUS	Sound Surveillance System	A series of hydrophones placed on the ocean floor to detect the noise emitted by Soviet submarines.
SSBN	Submarine, Ballistic, Nuclear	A submarine designed to carry ballistic missiles with nuclear warheads. The submarine itself is also nuclear powered (there are no diesel powered ballistic boats in the game).
SSGN	Submarine, Cruise Missile	Submarines with the primary task of cruise missile delivery.
SSN	Submarine, Nuclear	A nuclear-powered submarine.
SSK	Submarine, Diesel	A diesel-powered submarine.
TF	Task Force	A unit or formation of surface ships and up to one submarine counter.
TA	Towed Array	A very sensitive type of SONAR towed behind a ship or submarine.

“Fleet actions occur when some serious situation on land will eventually be intolerable if the weaker fleet does not come out to fight.”

- Capt. Wayne Hughes, USN, “FLEET TACTICS”, 1986

1.0 INTRODUCTION

First Time Player Notes

Welcome to Blue Water Navy!

The rules are individually not particularly complex and simply working through each section once or twice as you perform the relevant activity should be enough to get you playing. Once you are up and running, the game is designed to be playable directly from the quick play sheets with only occasional reference to relevant sections of the rule book.

The main complexity in the game really is the number of options the game presents you with – where do you attack? And with what forces? What about the cards? How about nuclear weapons? – the game has a great deal of options available, but in a single play through you will not get to see them all. The best advice is to have a plan each turn for what you want to achieve.

It is also easy to get 'tunnel vision' and concentrate on killing a particular target to the detriment of your overall plan – know when you've lost a fight and move on to somewhere you can win instead.

Modern combat is extremely deadly – do not be surprised to be taking off a large number of units every turn. An air strike can and will go wrong from time to time with awful casualties. If it's flying and fighting it might not come back, so be prepared to lose your toys when they fight.

Always remember the game is about the convoys – if they get through, NATO wins the war.

Blue Water Navy simulates a hypothetical 1980's war between NATO and the Soviet Union. The game takes place principally at sea and in the air, with ground operations represented in an abstract fashion. In the event of such a conflict, NATO convoys would form up in the United States and head to Europe and the Mediterranean carrying supplies and reinforcements. Their arrival would be of the utmost importance to the continued survival of NATO. The Soviets would attempt to interdict these convoys with their submarine fleet and their long-range bombers armed with stand-off anti-ship missiles while being supported by the Soviet surface fleet, simultaneously the Soviets would seek to conduct amphibious operations against the NATO flanks (Norway and Denmark).

1.1 The Sides

Players assume control of either the NATO forces or the Soviet forces. The background colour of each counter indicates side (NATO or Soviet), whilst the flag shows nationality.

NATO FORCES

United States



Spain



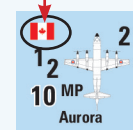
Netherlands



France



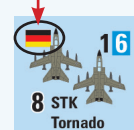
Canada



Italy



Germany



Norway



Great Britain



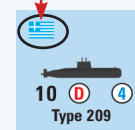
Portugal



Turkey



Greece



Denmark



SOVIET FORCES

Soviet



Cuba



Libya



1.2 Game Components

- 1 map
- 4 counter sheets
- 2 decks of 55 playing cards (110 cards)
- 8 double-sided player aids
- 1 single-sided OPS track for smaller scenarios – single sheet card
- 1 scenario book
- 1 rule book
- 10 ten-sided dice
- 1 box and lid set

2.0 COMBAT UNITS

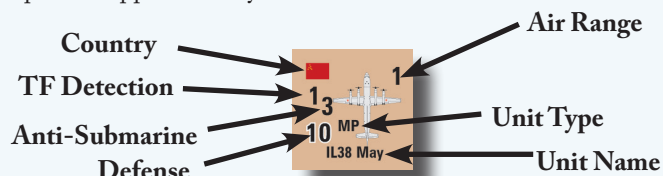
Combat Units for both sides are classified as either **Air Units**, **Submarines Units**, or **Surface Units**.

2.1 Air Units

Air Units are either Maritime Patrol (MP), Strike (STK) or Fighters (FTR). Additionally, the Soviet player has Tanker Aircraft (TKR).

2.1.1 Maritime Patrol (MP)

Maritime Patrol units are aircraft designed to operate for long durations over water conducting Anti-Submarine Warfare (ASW) and Anti-Surface Warfare (ASuW). They are one of the primary methods for detection of surface Task Forces. MP units represent approximately 15 aircraft.



Country – The nation the unit belongs to.

Task Force Detection – The number of dice used for detecting Task Forces.

Anti-Submarine Value – The number of dice used for attacking submarines.

Defense – The number required by an enemy attack die to cause a step loss to the unit.

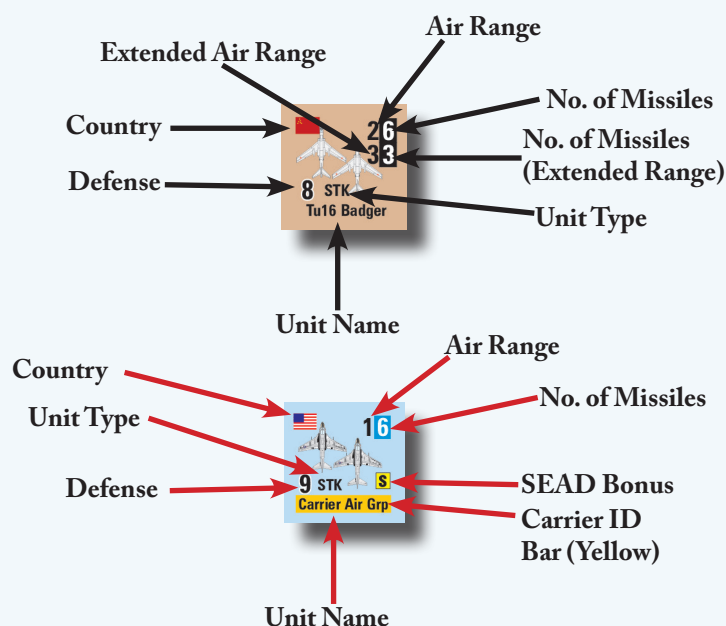
Unit Type – MP.

Unit Name – Historical identification.

Air Range – The unit's movement allowance in Sea Zones.

2.1.2 Strike Aircraft (STK)

Strike units are aircraft designed to fly attack missions either with bombs or missiles. Strike units represent 24-30 aircraft plus any supporting aircraft (Recon, Electronic Warfare, Tanker etc.)



Country – The nation the unit belongs to.

Unit Type – STK.

Defense – The number required by an enemy attack die to cause a step loss to the unit.

Unit Name – Historical identification.

Air Range – The unit's movement allowance in Sea Zones.

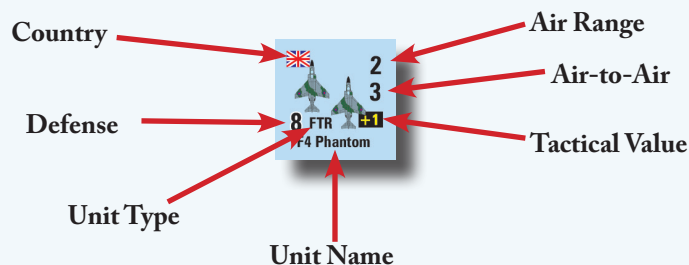
Number of Missiles – The number of weapons carried and their type indicated by the color of the text and the background color.

SEAD Bonus – US Carrier Air Groups receive a bonus when attacking enemy SAM sites, which represents EA-6 Prowler aircraft.

Carrier ID Bar – The color shows the carrier the unit is based on.

2.1.3 Fighter Aircraft (FTR)

Fighter units are aircraft designed for air to air combat missions. Fighter units represent 24-30 aircraft plus any supporting aircraft (Recon, Electronic Warfare, Tanker etc.)



Country – The nation the unit belongs to.

Unit Type – FTR.

Defense – The number required by an enemy attack die to cause a step loss to the unit.

Unit Name – Historical identification.

Air Range– The unit's movement allowance in Sea Zones.

Air to Air– The number of dice used for air to air combat.

Tactical– This number is added to each air to air die rolled.

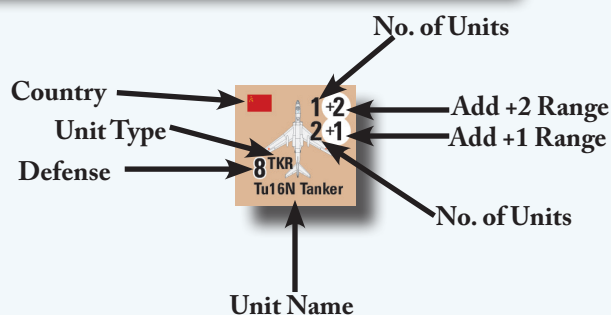
Carrier ID Bar – The color shows the carrier the unit is based on.

2.1.4 Soviet Tanker Aircraft (TKR)

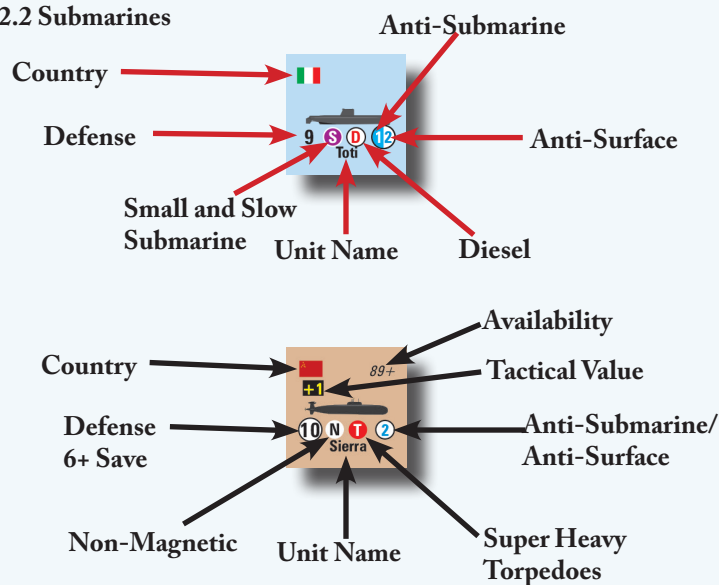
The Soviet player begins with 2 Tanker Aircraft added to their force to increase the ranges of Strike aircraft. Further details on refueling can be found at [13.1]

Number of units & Tanker Range-

This is the additional range a tanker can give to Strike aircraft. A full-strength 2-step tanker can add +1 range to 2 units or +2 range to one unit.



2.2 Submarines



Country – The nation the unit belongs to.

Tactical – This number, if present, is added to each anti-submarine or anti-surface die roll. (when not present the unit adds nothing)

Defense – The number required by an enemy attack die to cause a step loss to the unit. **If within a white circle**, the unit has a special 6+ save against each step loss it takes due to superior speed, stealth or countermeasures.

Unit Name – Historical identification.

Diesel (D) – A diesel submarine, slower than a nuclear submarine.

Non-Magnetic (N) Specially treated or constructed submarine hulls making detection from the air more difficult and preventing mines from detonating.

Small and Slow Submarine (S) –The submarine is small and gets a bonus against minefields but is slow and can only move OR attack.

Super Heavy Torpedoes (T) – Used by the Soviets to sink US aircraft carriers.

Availability - Denotes availability of some units by year.

Missile Type - The missile type & number of weapons carried.

Anti-Submarine/Anti-Surface - The number of dice used for anti-submarine warfare and anti-surface attacks.

Submarines are either Nuclear-Powered (**SSN/SSGN/SSBN**) or Diesel-Powered (**SSK**). Diesel-powered submarines are indicated by the (D) symbol.

Some Soviet nuclear submarines are **SSBNs** ("Submarine, Ballistic missile, Nuclear powered", also known as "Boomers") which carry nuclear ballistic missiles. These are indicated by their Payload value (yellow number in a red triangle)

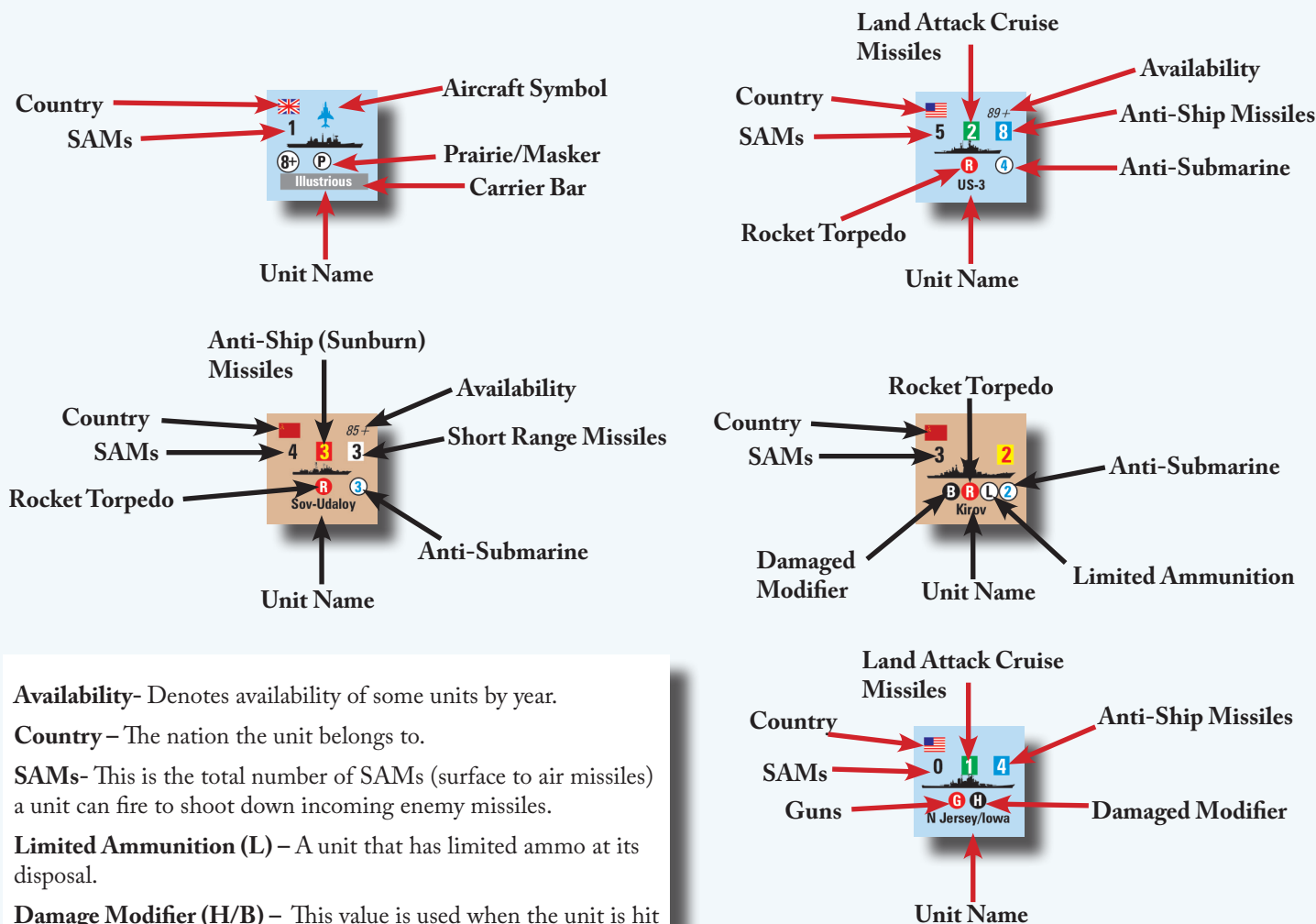


A unit represents three submarines of that class except for diesel-powered submarines with an **Anti-Surface** value of 4 which represents six submarines of that class.

2.3 Surface units

Surface Units represent each sides fleet of warships, Amphibious Landing Ships, aircraft carriers and helicopter carriers. Each unit represents either a single capital ship or a group of between 6-10 warships operating together. A unit with an Aircraft symbol close to its name is an **aircraft carrier**.

Note: A unit with a helicopter symbol is a helicopter carrier and has no Air Unit – it contributes to ASW only. All carriers have some detection capability due to AEW aircraft.



Availability- Denotes availability of some units by year.

Country – The nation the unit belongs to.

SAMs- This is the total number of SAMs (surface to air missiles) a unit can fire to shoot down incoming enemy missiles.

Limited Ammunition (L) – A unit that has limited ammo at its disposal.

Damage Modifier (H/B) – This value is used when the unit is hit to determine damage sustained.

Rocket Torpedo (R) – A unit equipped with rocket torpedoes.

Guns (G) – This is a unit with large caliber guns.

Slow Ship (S) – Units marked 'Slow' have a reduced movement allowance and are easier for submarines to attack.

Prairie/Masker (P) – Units receive a bonus against torpedo attacks – it is a type of acoustic masking system.

Only NATO has this technology.

Land Attack Cruise Missiles – The number of dice the unit can use to attack land targets with cruise missiles.

Anti-Ship Missile – The number of anti-ship missiles carried by the unit. The color of the text shows the type of missile.

Anti-Submarine – This is the unit's value when attacking submarine units, or when detecting submarines attacking their Task Force.

Capital ships – All aircraft carriers and any ship with limited ammunition (L) or a damage code (B) or (H) is a capital ship, except for the US amphibians – which are amphibious units.

Ships always operate in **Task Forces** [11.0] which are 'filled' by placing Surface Units into the relevant Task Force box on the map. The ships remain in the Task Force box and the Task Force marker moves on the map.

Note: The Task Force boxes are private and cannot be inspected by the opposing player. Detection of a Task Force will reveal the ships within.

2.4 Other Units

Additional units used in play include: RORSAT (Soviet Reconnaissance Satellites), SAM sites and Convoys or Amphibious units.

Detection dice vs
Task Forces



Number of dice against strike
aircraft using bombs



Detection dice vs
submarines

Number of dice
vs cruise missiles

Note: US Amphibs have the Helicopter symbol indicating a helicopter carrier. (I assume some of the marines' helicopters are on ASW Duty).

2.5 Damage and Step Loss

Damage dealt to units is called a Step Loss. Most units have two sides referred to as 'steps', when a unit takes a 'Step Loss' flip the counter to its weaker side. If already flipped, remove the counter to the Destroyed box. If there is no weaker side, the unit is destroyed by a single step loss.

2.6 Step Combination

You may (not must) freely combine two one-step units of the exact same type and activation status ('Fresh' or 'Spent') in the same location together to make a single full-strength unit at any time. Units combining in this way may activate normally.

2.7 Missile Types

Submarine Units and Surface Units are equipped with an array of missile weapons, shown here on the Missile Types Table. For ease of play missiles are referred to using the color of the missile background eg: *Green is a land attack cruise missile, while yellow is a Shipwreck missile.*

Missile Types	
NATO Types	
2	Land Attack Cruise Missiles
2	Anti-Ship Missiles
Soviet Types	
2	Anti-Ship (Sunburn) Missiles
2	Anti-Ship (Shipwreck) Missiles
2	Short Range Anti-Ship Missiles
2	Long Range Missiles
2	Land Attack Cruise Missiles

2.8 Stacking

All units on a side may stack and operate together without penalty. There are no stacking restrictions and units of both sides can freely exist within the same Sea Zone which represents a large area.

Exception: A Task Force may only have one Submarine Unit with it as an escort, and it must be a nuclear submarine (not an SSBN symbol), not a diesel (D).

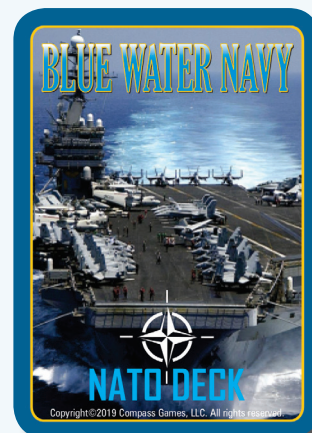
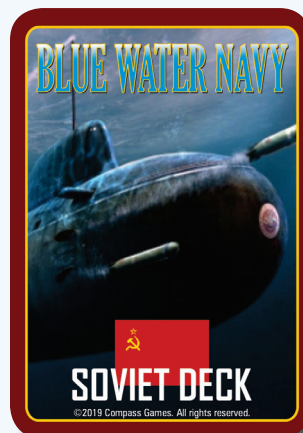


Design note: Diesel submarines are too slow to escort surface ships, even slow ones.

There are no other stacking restrictions – any number of submarines, ships and or Air Units can simultaneously exist in the same location.

3.0 CARDS

Each side has a deck of cards which drive the game. Each card has a numeric value referred to as 'Operations Points' in the top left corner (also 'OPS') along with one or two events. The top event on a card can only be used if the card is played for Operations Points (or by 16.6.5), the bottom event can be played whenever the conditions specified are met. Two cards are played each turn for Operations Points by each player.



Once a card is played it is moved to your own discard pile which will be reshuffled if you need to draw new cards and have no remaining cards in the deck.

A card with '**remove**' specified within the event text is removed from the game after the event is played.

The top event on each card has a cost to play indicated by the right hand value of the card - this must be paid from the Operations Points granted by play of the card if the player wishes to activate that event.

The bottom event on each card is played in response to something happening and is known as a '**reaction**'. Any number of reactions can be used as and when applicable, without any limitations on the number of cards expended and with no cost in ops points.

A player will always have three cards.

Whenever a player drops to only two cards, draw another card.

4.0 DICE

All dice are ten-sided. Any 'zeros' are read as 'tens'.

4.1 Die Roll Modifiers (DRM)

Throughout play various events and actions will involve Die Roll Modifiers (DRM) that when combined with the initial die roll result, give you the final result. A DRM can be positive or negative.

4.2 Natural Rolls

A 'natural roll' is a die result before modifiers are applied. Thus, a natural '10' can only be achieved through the roll of a '10' regardless of DRMs. In this game, a natural roll usually generates a special result and is preceded with 'N' on the combat tables.

5.0 UNITS & STATUS (FRESH or SPENT)

Each unit (counter) is considered '**Fresh**' until it completes an action which makes it '**Spent**' or '**On Patrol**'. Use the relevant 'Spent' and 'On Patrol' markers to signify unit status.



Note: The following are exceptions to this:

- Task Forces are never marked 'Spent'.
- Surface units are never marked spent
- Fighter units on CAP are never marked "Spent" by flying CAP alone.

A unit which is not marked 'Spent' or 'On Patrol' is 'Fresh' and is eligible to perform any legal action [16.1].

While marked 'Spent' a unit can take no action but may be able to fly limited CAP if a Fighter unit.

While marked 'On Patrol' a unit can react to enemy movement as per the On Patrol rules or remove its On Patrol marker during the turn if it is an Air Unit. Submarines can only remove their On Patrol marker at the end of the turn.

6.0 GAME MAP

6.1 Game Map

The map covers the area from the United States to the western Soviet Union and includes the Mediterranean Sea. Sea Zones are approximately 500 nautical miles wide.



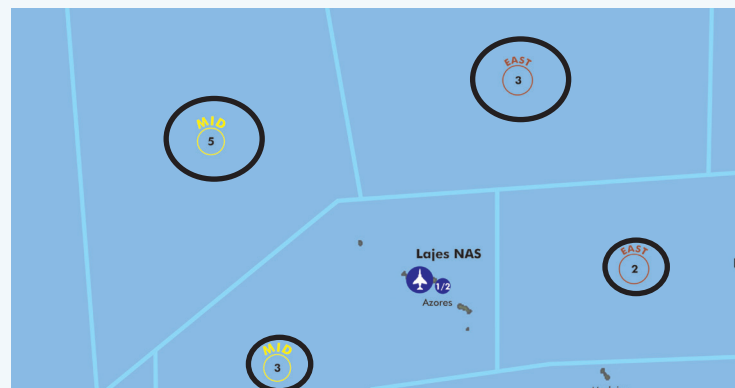
6.2 Sea Zones

The map is divided into Sea Zones to regulate movement and combat. Units can move between Sea Zones without restriction with the following exceptions:

- The Arctic zone is restricted to nuclear submarines only - no (D) submarines [6.4].
- Bad Weather slows surface movement [12.1] and has many combat related effects.
- NATO may react to Soviet units moving across a Hydrophone Barrier [6.8] and similarly Soviet units may react to NATO units entering the Bastion.

6.3 Weather Zones

All Sea Zones except for the Arctic [6.4] are grouped into one of five "**Weather Zones**", West, Mid, East, Med and North and identified by a Weather Zone box and number (e.g. West 6, North 3-4). These identifiers are used for scenario set-ups and other game functions such as determining the location of Bad Weather [10.0]. Some Sea Zones share the same Weather Zone identifier (e.g. West 1, North 9-10). If Bad Weather is rolled for a shared zone identifier, place Bad Weather markers in both Sea Zones.



6.4 The Arctic

This special “under ice” Sea Zone can only be entered by nuclear submarines (any submarine without a (D) symbol).

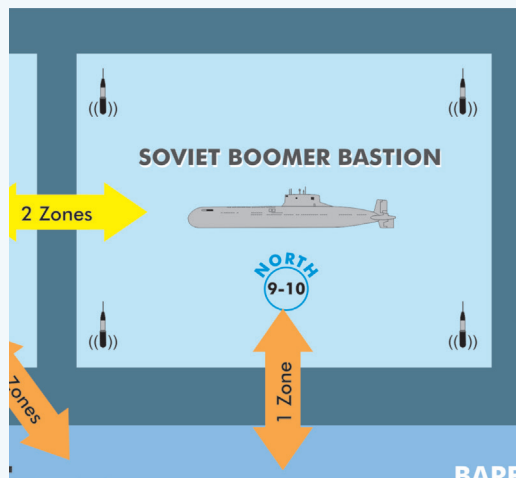
The number of movement points needed to enter or leave the Arctic is indicated by the arrows on the North side of the map. Note that only NATO submarines may enter or exit the Arctic via the Labrador Sea east of Canada (Sea Zone Mid 10) – see the note on the map.

The Arctic does not have a Weather Zone identifier and never suffers the effects of Bad Weather. But the constant shifting and grating together of ice makes underwater combat there difficult due to the noise generated by this process.

Design note: In addition to the protection provided to Soviet SSBNs by the Arctic, some Operation Events grant bonuses when SSBNs operate under the ice.

6.5 Soviet “Boomer” Bastion

This special Sea Zone represents the area north of the Barents Sea where the Soviets created a fortified area to protect their SSBNs. It contains a Hydrophone System (“CAESAR”) to help detect NATO submarines. It is mined and has its own defenses - corvettes and air patrols. It has the following game effects:



Movement: The Soviet “Boomer” Bastion can be entered or exited by all unit types at the movement point cost indicated by the arrows on the map.

Combat:

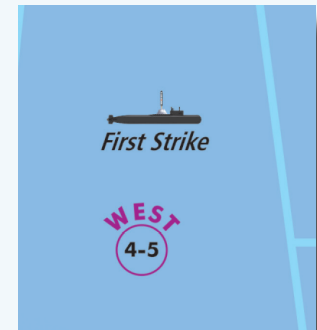
- Hydrophone Reaction applies [6.8] to any NATO submarine entering the Bastion.
- NATO submarines rolling one or more natural ‘1’ result(s) when performing Anti-Submarine Warfare take a single step loss. (representing the Bastion’s defenses)

6.6 First Strike Zones

Several ‘First Strike’ zones are marked on the map. Some are for any SSBN and some are only for non-Yankee class SSBN. A submarine only counts when required for a First Strike check if it is in a zone where it can be counted.

Example: A Yankee SSBN in a non-Yankee zone would be ignored.

Design note: Historically, the USSR ran patrols in these regions which gave them a First Strike capability in that they could launch and hit the East coast of the USA in a few minutes of flight time. The downside is that they are close to the USA and are prime targets for destruction by NATO anti-submarine forces.



6.7 First Strike Evasion

Before a NATO unit rolls Anti-Submarine Warfare dice in or adjacent to a First Strike Zone, the Soviet player **may** choose to roll a die for each SSBN within the zone.

A 6+ is required if the SSBN is ‘Fresh’ and 8+ if it is ‘Spent’. If successful it may be immediately moved to an adjacent Sea Zone, along with a single other non-SSBN, non-diesel (D) submarine (its escort) both units are then marked ‘Spent’ if they are not already.

NATO must still search with the active unit even if there is no enemy unit left to find.

An unsuccessful evasion has no effect.

Design note: This rule accounts for the ‘gods eye view’ that players have of the map and makes SSBNs slippery. Remember: Always escort your First Strike submarines!

6.8 Hydrophone Barriers

During the Cold War, NATO deployed Hydrophone Sound Surveillance Systems (“SOSUS”) at Atlantic Ocean chokepoints to track Soviet submarines. The table below shows their locations as well as their ASW assets:

Barrier	Sea Zone Borders	Local ASW Asset
GIUK Gap	East 6, East 9-10, North 3-4	No
English Channel	East 4	[Yes 2 dice Helicopters unless Bad Weather.]
Strait of Gibraltar	Med 1-2	[Yes 2 dice Helicopters unless Bad Weather.] (See #3, top of page 11)

When an enemy submarine crosses a Hydrophone Barrier shown on the map along a zone’s border, one ‘Fresh’ or ‘On-Patrol’ friendly unit in the Sea zone being entered or exited via that border may interrupt the movement for each submarine moving across that border using one of the following options:

R#12

Note that Bad weather applies if it is in the zone the submarine is moving INTO, not out of

R#3

1. Roll a single die on the Anti-Submarine Warfare Table [17.8] against the unit entering the zone, adding the Tactical Value of the Fresh friendly unit.

Note: There is no negative effect on the friendly unit.

2. Roll a normal Anti-Submarine Warfare attack [17.8] against the unit entering the zone as though the Fresh friendly unit was attacking it (and only it). Roll one additional die and then mark the friendly unit 'Spent'.

3. The English Channel and Straits of Gibraltar have a permanent local ASW asset rolling two dice as though it were a Maritime Patrol unit (applying weather and (N) normally from the Anti-Submarine Warfare Table [17.8]). This roll is used instead of one of the other two options listed above. In Bad Weather, this is reduced to zero dice.

Examples: *A Soviet submarine moves from North 7-8 over the SOSUS line into and through East 6. The NATO player can activate a Fresh unit in East 6 and roll Anti-Submarine Warfare with +1 die. Alternatively, NATO can roll a single die against the Soviet submarine and does not mark the detecting friendly unit 'Spent'. In this way NATO can claim a free die against every Soviet submarine crossing the SOSUS line so long as Fresh units are available.*

If the Soviet submarine moved east across the SOSUS line heading back home, the submarine still crosses the Hydrophone Barrier and the same process would apply.

6.9 Land Areas

These are areas of the map that contain ports and/or airfields, referred to as "Facilities". The number of airfields or ports shown on the map is for interest only – for game purposes a Land Area has a maximum of **one** port and/or **one** airfield facility (or a combination of both, see [6.10.3]).

Example: *The Kola Peninsula has many airfields shown but for game purposes it is treated as a single airfield facility and a single port adjacent to the North 9-10 Sea Zone.*

The airfield and port symbols are used to show which Sea Zones the facility is adjacent to. Any symbol or airfield/port box touching a Sea Zone indicates adjacency. Exception: The Kola Ports are only adjacent to North 9-10.

Example: *The UK has two facilities. The airfield is adjacent to North 5-6 because Lossiemouth touches the Sea Zone, North 3-4 because Leuchars touches the Sea Zone, East 6 because of Stornoway, and East 4 because the facility box itself touches that Sea Zone.*

The location of port symbols on the map defines port adjacency –

Example: *Keflavik NAS (Iceland) is adjacent to North 7-8 and 9-10. But its port is only adjacent to East 9-10 because the port symbol touches no other Sea Zone.*

Units can move from a facility to any adjacent Sea Zone and vice-versa, although naval units are subject to movement rules for entering and leaving ports [12.3].

Example: *Units on the UK airfields can move to North 3-4, North 5-6, East 6 or East 4 because these Sea Zones are adjacent to the airfields.*

When a Sea Zone experiences **Bad Weather**, any Land Area adjacent to that Sea Zone experiences Bad Weather as well.

Example: *The UK airfields will experience Bad Weather if any of the 4 adjacent Sea Zones has Bad Weather.*

6.10 Facilities

There are three types of facilities – **Ports**, **Airfield**, and **Combined Facilities**.

6.10.1 Ports

Ports represent significant naval shore facilities used to form Task Forces, receive NATO convoys, repair capital ships and re-arm naval units. There are two types of ports – **Major** or **Minor**. The capability of each type is noted below.



Minor Port

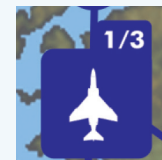


Major Port

Type	Capability
Minor port	May rearm one friendly Submarine Unit per turn and may only rearm Surface Units of own nationality.
Major Port	May rearm any number of friendly Submarine Units and may rearm any number of friendly Surface Units. Also may repair capital ships which are damaged.

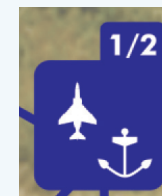
6.10.2 Airfields

Airfields can accommodate and operate an unlimited number of ground-based aircraft.



6.10.3 Combined Facility

Ports and airfields combined into a single facility, they have a single Damage box and so if damaged, both the airfield and the port are damaged simultaneously to the same level of damage.



Example: *Esbjerg in Denmark is the same facility as Kiel and Schleswig which is a combined facility.*

6.10.4 Facility Damage

Each facility type has a Damage box which shows the number of hits required to cause 'Light' and 'Heavy' damage. If a Land Area has separate ports and airfields, it will have separate Damage boxes, while a Land Area with a combined facility has a single Damage box that affects both facility capabilities equally

Example 1: *The United Kingdom has an airfield facility in the North and a port facility in the South. Each facility has a Damage box (1/3 and 2/4 respectively) and is damaged independently.*

Example 2: *NS Rota in Spain is a combined facility with a 1/2 Damage box.*

When a facility is damaged, place the number of hits scored by the attacking player on the facility (along with any pre-existing hits) to a maximum of 5. Then evaluate the damage level – if the number of hits is equal or higher than the first number, the facility is lightly damaged. If the number of hits is equal or higher than the second number, the facility is heavily damaged – place the correct Damage Level marker and then check the

Damage Effect Table and apply the results.

Damage is cumulative – apply both light and heavy damage results to a heavily damaged facility.

R#18

Damage	Effect of Damage
Airfield Light	No Air Units may fly, return to base any Air units that are On Patrol and mark them 'Spent'. Also applies a CAP modifier [17.7]
Airfield Heavy	No Air Units may fly. All CAP is reduced to one die per unit. Do not remove 'Spent' markers on Air Units when ships move or at turn end – the airfield has lost the ability to refuel and maintain aircraft
Port Light	• Causes +1 hit to Convoys and Amphibs arriving or leaving for delay • +1 additional Operations Point to activate a Submarine Unit at this port (2 OPS). • Mine clearance penalty of -2
Port Heavy	• Causes +2 hits to Convoys and Amphibs arriving or leaving (instead of +1) • Forming a Task Force at this port requires +1 Operations Points.

Design note: *This means that a facility which requires 3 hits to be 'Heavily' damaged, such as the UK airfields can have excess hits placed on it to keep it suppressed for longer, and that a large facility which is 'Lightly' damaged on two hits will recover faster.*

Example: *The RAF airfield in Scotland is hit. They take one damage and so a Hit marker is placed. Since the Damage box shows 1/3, one hit is enough to inflict light damage, but it would require three hits to cause heavy damage. A Light Damage marker is placed.*

6.10.5 Facility Repairs

Each turn the number of hits on each damaged facility will decrease by two (one per Operations card played), and the owning player can additionally take a Repair action to try and speed this up [16.5.1]. When the number of hits is less than the 'Heavy damage' value, the facility goes from 'Heavy damage' to 'Light damage'. When the number of hits drops below that required to cause 'Light damage', the facility is no longer damaged.

6.10.6 Facility Capture

Several NATO facilities can be captured by the Soviets during the game by Amphibious Landing or through advancement of the War Tracks [19.0 and 20.0].

6.11 Map Overflight Arrows

Several arrows show an alternate route a unit may take to reach a Sea Zone. The arrow explanation text will show either "+X" where X is the additional cost in movement to use the arrow or "=X" where X is the total movement cost to use the arrow.

Example 1: *An arrow starts in North 9-10 and then overflies the Arctic. For an additional 2 zones of movement, the unit can fly from North 9-10 to East 9-10. This may allow avoidance of NATO fighters. The total distance moved is 3 (one to the Barents Sea, plus two to follow the arrow).*

Example 2: *From the Black Sea bases, Air Units may fly directly to the Baltic Sea for a total movement cost of 3 zones or to Med 7-8 for 1 zone (but overflying Turkey – [21.1]) or to Med 7-8 for a total movement cost of 4 zones by overflying Iran, Iraq and Syria – which is a safer but much longer route.*

6.12 Measuring Range

For movement purposes a Land Area is one zone away from all adjacent Sea Zones, however when a facility is being attacked the attacking units only need to be able to reach an adjacent Sea Zone.

Example: *Flying from the Kola Peninsula bases to the Barents Sea is 1 zone of movement, but to attack the Kola bases, NATO need only be able to trace adjacent to them – flying to the Barents Sea is therefore good enough for attacking purposes.*

An "island" Land Area (e.g. Puerto Rico) situated entirely within a Sea Zone is at range zero to its surrounding Sea Zone and range one to neighboring Sea Zones.

6.13 Game Tracks and Holding Boxes

The map has various tracks to hold game information markers, and several holding boxes for Task Forces and Destroyed Units. Task Force composition is indicated by placing Surface Units into the relevant Task Force box on the map. Task Force composition is private information to the owning player. If a Task Force is detected the contents will be placed on the map in accordance with the detection rules.

7.0 DETERMINING VICTORY

The scenarios each specify their own Victory Conditions, but for the campaign game:

The Soviets must be able to count four 'Hammer and Sickle' icons on locations overrun by their forces on the War and Invasion Tracks.

Example: *At the end of the game, the Soviet player has troops in the Oslo box of the Norway Invasion Track (1 hammer and sickle); and has an army in the 'Ruhr Cities' box of the Europe North War Track (2 hammer and sickles) – no other occupied boxes contain hammer and sickles. The Soviet player has lost.*

Soviet forces automatically advance on the War Tracks each turn (Germany, Italy, Turkey, etc.)— this is slowed or stopped entirely by the NATO player delivering Convoys without too many hits – a Convoy with up to two hits prevents an advance and a Convoy with up to 4 hits counts as a ‘partial Convoy’, two of which combine to prevent one advance. (one partial Convoy alone does nothing).

NATO may strike Soviet forces to slow the advance on the Europe South War Track (Italy, Turkey) by attacking the Soviet Troops icon located in Yugoslavia / Greece [17.24].

The Invasion Tracks (Norway and Denmark) may advance depending on the delivery of troops by Amphibious units. Both sides have these units and delivered troops cancel each other out. For the Soviet player to advance, at the end of the turn there must be Soviet Troops markers on the track equal to the Troops value of the box they are advancing into. NATO may halt an advance on either Invasion Track by striking the advancing Soviet troops [17.24].

In addition to the ‘Hammer and Sickle’ at the end of each Invasion Track (representing the surrender of Denmark or Norway) each track also provides additional benefits to the Soviet player – the Norwegian bases can be occupied to provide forward airfields and the Danish Strait can be de-mined, allowing Soviet fast attack units to flood out into the English Channel [21.2.1].

Note: Moving into a Victory Point space using battlefield nuclear weapons [22.1] reduces the value to 1 hammer and sickle. It does not further reduce a space already worth only 1 hammer and sickle.

8.0 GAME SETUP

See the scenario book for full setup instructions.

9.0 TURN SEQUENCE

The game lasts for a maximum of twelve turns, each turn representing 2 days.

1. Convoy Placement Phase

Place Convoy units appearing this turn in any US port (not Canada).

2. Reinforcement Phase

Receive all forces entering the game this turn in any friendly port of that nationality.

Soviet RORSATs appear in the Tyuratam box

3. Operations Phase

- The Soviet player selects a card to play for its Operations Points and records the number on the Operations Point Record Track with the Soviet marker.
- The NATO player selects a card to play for its Operations Points and records the number on the Operations Point

Record Track with the NATO marker.

- The Soviet player plays the first Operations Point to perform an action or Passes [16.0].
- The NATO player plays an Operations Point to perform an action or Passes [16.0].
- Continue alternating through c. and d. until no Operations Points remain.

Look for events on the Operations Point Record Track: Fast, Repair, Ships, and Next Day. These trigger when both players’ Operations Points marker reaches or passes the box and takes effect immediately after the current action is resolved.

4. End of Turn Phase

- Fighters on Interception missions and ‘On Patrol’ MP units may return to base.

If a zone contains Fighter units of both sides, all Air Units return to base.

- Remove all ‘Spent’ markers (except for Air Units at a heavily damaged base).

Choose to remove ‘On Patrol’ markers on Submarine Units, or to retain them. Submarines with ‘On Patrol’ markers can only react to enemy units.

c. Invasions:

- Resolve the Invasion War Tracks (Denmark and Norway).
- Increase the Danish Strait Mining level if no Soviet troops ever landed in Denmark.
- Roll to clear the Danish Strait of mines if the War marker is in a relevant box.
- Select a bonus if Norway has surrendered.
- Remove submarines if required by the Invasion Tracks.

d. Calculate War Track changes for Convoys and Soviet advances, then remove submarines, capture facilities and resolve NATO nuclear weapon usage if required by the War Tracks. Remove 3 hits on the Europe South War Track and the Norway and Denmark Invasion Tracks.

- If the current turn has a First Strike icon, award the Soviet player 1 First Strike Point if at least 6 SSBN Payload value are in First Strike zones (ignore the Arctic and the Bastion).
- Remove all Weather markers and roll for weather in each Weather Zone on the map.
- Choose to hold up to two cards (NATO); one card (Soviet) and discard all other cards Then, draw up to 5 cards in-hand. If you held cards, you’ll draw less new cards.

Note: Holding cards is not mandatory.

- Move the Turn marker – The game ends at the end of turn 12.

- Remove one First Strike Point from the ASAT System box on a die roll of 6+.

9.1 Detailed Turn Sequence

9.1.1 Convoy Placement Phase

During each turn, NATO will receive one Convoy and may receive a Mediterranean Convoy. The NATO player may then select from any of the counter types available (except the Fast Convoy, which enters via event only). All Convoys are then placed on the map in a port in the United States.

9.1.2 Reinforcement Phase

Depending on the set-up chosen, each side has several units that will become available during the game. These include Amphibious units for both sides, RORSATs for the Soviets, and Carrier Battle Groups for NATO.

During the Reinforcement Phase the units arriving are placed on the map. All reinforcements arrive in any port belonging to the owning nation. Soviet RORSATs are placed in the Tyuratam box.

9.1.3. Operations Phase

The Soviet player selects a card to play for Operations Points and records the number on the Operations Point Record Track with the Soviet marker.

The NATO player selects a card to play for Operations Points and records the number on the Operations Point Record Track with the NATO marker.

This means the NATO player can see what the Soviet player plays before deciding on a card play.

The Soviet player plays the first Operations Point to perform one action, which is selected from the Actions Chart [16.1]. Any action may be selected including those from the 'use when active – not an action' portion of the chart.

Note: These actions may be used in addition and do not cost Operations Points. Once the Soviet player chooses and completes an action, play passes to the NATO player.

The NATO player then plays an Operations Point to perform an action and follows the same steps as the Soviet player.

Continue alternating between players performing one action each until no Operations Points remain. A player may also **Pass** when they have **less** Operations Points than their opponent. They do nothing, and the other player plays another action, play then continues normally.

Look for events on the Operations Points Record Track: **Fast**, **Repair**, **Ships**, and **Next Day**. These trigger once both player's Operations Points markers reaches or passes the box and takes effect immediately after the current action is resolved. They are described below:

Fast: If it is not Day Two: Fast Task Forces may move one Sea Zone (they may not perform ASW). If it is Day Two, ignore this event. Resolve as per the Ships event, ignoring steps 2 and 6, and applying only to Fast TFs and zones they occupy.

Repair: Remove one hit from each damaged facility on the map and then check to see if the facility is now lightly damaged or no longer damaged [6.10.5].

Ships:

1. Move Task Forces.

Both sides move all Task Forces according to the movement rules [12.0].

The Soviet player moves all Task Forces **first**, followed by NATO within each detection category.

Task Forces may prepare for an Amphibious Landing by marking themselves with the 'Amphibious Landing' marker. For each that does, roll a free Task Force Detection roll [17.10].

Task Forces already marked with an 'Amphibious Landing' marker may resolve their landing, and then remove the marker. If they do so they may not move.

If a Task Force moves into or from a zone with a friendly nuclear submarine which is not Spent or On Patrol, the submarine may join the Task Force so long as it is the only submarine with the Task Force.

Instead of moving a Task Force, it may conduct Anti-Submarine Warfare in its current zone [17.9].

Units marked as 'On Patrol' may react to enemy Task Force movement. [16.2.1, 16.3.1, 16.3.2]

Air Units marked 'On Patrol' may move with a Task Force that just moved so long as they have the range to operate there (i.e. they must remain within range of their base).

2. Degrade existing Task Force detections that were not placed in step 1 (if they have an Amphibious Landing marker, do not degrade the Detection level).

All Good detections are reduced to Poor detections and all Poor detections are removed. See [11.1] for the effect of downgrading Task Force Detection levels.

3. Roll for Task Force Detection with each RORSAT on the map which shares a Sea Zone with one or more enemy Task Force(s).

4. Resolve Fighter vs Fighter combat due to the movement of Task Forces [17.6].

5. Resolve Interception and CAP response to intrinsic detection units and then resolve their searches.

6. Remove 'Spent' markers from all non-MP Air Units (except for Air Units at a base with Heavy damage).

Next Day: Both players play their second (and last) card of the turn for its Operations Points, exactly as per 9.1.3, adding the number of Operations Points still unspent to the total available to you.

Advance the Day marker to 'Day Two' then continue play.

Note: On Day Two the Fast and Next Day events are ignored and have no effect.

9.1.4. End of Turn Phase

- Fighters on Interception missions and MP units that are 'On Patrol' MP may return to base.

If a zone contains Fighter units of both sides, then all Air Units in the zone must return to base

- Remove all 'Spent' markers (except for Air Units at a 'heavily' damaged airfield).

Choose to remove 'On Patrol' markers on Submarine Units, or to retain them. Submarines with 'On Patrol' markers can only react to enemy units.

c. Invasions:

1. Resolve the Invasion War Tracks (Denmark and Norway).
2. Increase the Danish Strait Mining level if no Soviet troops ever landed in Denmark.
3. Roll to clear the Danish Strait of mines if the Soviets occupy a relevant box on the track.
4. Select a bonus if Norway has surrendered.
5. Remove submarines if required by the Invasion Tracks.

d. Calculate War Track changes for Convoys and Soviet advances, then remove submarines, capture facilities and resolve NATO nuclear weapon usage if required by the War Tracks. Remove 3 hits on the Europe South War Track and the Norway and Denmark Invasion Tracks.

e. If the current turn box has a First Strike icon, award the Soviet player 1 First Strike Point if at least 6 SSBN Payload value of 6 are in First Strike zones (ignore the Arctic and the Bastion).

f. Remove all Bad Weather markers and roll for weather in each Weather Zone on the map.

g. Choose to hold up to two cards (NATO); one card (Soviet) and discard all other cards.

Draw up to 5 cards in-hand. If you held cards, you will draw less cards.

Holding cards is not mandatory.

h. Advance the Turn marker – **The game ends at the end of Turn 12.**

i. Remove one First Strike Point from the ASAT System box on a die roll of 6+.

10.0 DETERMINE WEATHER

To generate weather, resolve the following process at the end of each turn (and the beginning of Turn 1 as per the scenario setup instructions). First remove all Bad Weather markers from the map.

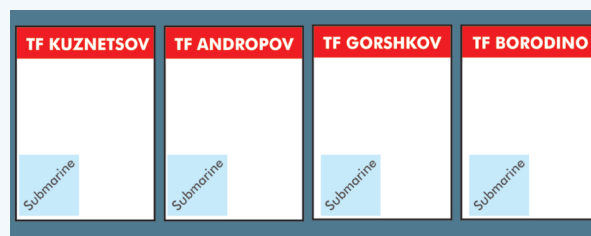
Then, roll one die for weather in each Weather Zone (West, Mid, East, North, Med) and use the appropriate table printed on the map. If the result is a number, roll that many dice and place Bad Weather in all Sea Zones with the matching number within that Weather Zone - some areas have more than one zone with the same number (eg: Barents Sea and the Bastion are both 9-10).

A Sea Zone can only have a Bad Weather effect once a turn. If a rolled zone already has a Bad Weather marker, ignore the dice which rolled the same number – that die roll generates no further weather effect. Land bordering or within a zone with Bad Weather also has Bad Weather. Some land is affected by multiple Sea Zones

Example: *The North Weather Zone is rolled for and the result is an 8 which is '2' – two dice are rolled – 7 and 8. The first Bad Weather marker is placed into North 7-8 to account for the 7. The 8 is ignored because it's a duplicate result.*

11.0 TASK FORCES

Task Forces are formed by placing Surface Units into the relevant Task Force box on the map. The ships remain in the Task Force box and the Task Force marker moves on the map. This helps reduce map clutter and makes tracking of units easier.



Soviet Task Force Boxes on the map



NATO Task Force Boxes on the map



Task Force Markers

Task Forces (TF) are made up of any number of Surface Unit counters and up to one non-SSBN nuclear submarine (without a (D)). Any number of available Convoy and Amphibious units can be attached to a Task Force. The force becomes **Slow** if any slow units (counters with an (S)) are part of the Task Force, otherwise it is considered **Fast**. Each TF must contain at least one Surface Unit.

Task Forces may remain at sea indefinitely. There is no supply limitation in terms of food, fuel etc. SAM ammunition is not tracked (except for units marked (L)). Anti-ship missiles and cruise missiles are tracked either as expended or not expended [13.9].

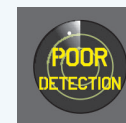
Whilst at sea a Task Force can merge with another Task Force, swap units between Task Forces in the same zone, detach units (such as Convoys arriving into a port) or create new Task Forces [16.4.1].

11.1 Task Force Detection

A Task Force has three possible states: **Undetected, Poor Detection, Good Detection.**

A unit marked as a Poor Detection which is detected again is upgraded to a Good Detection. Undetected Task Forces have their contents hidden from the enemy and cannot be attacked.

Poor detection Task Forces reveal the presence of Amphibious units and Convoys. Place all Convoy and Amphibious counters within the Task Force close to the Task Force marker on the map so that the enemy player can easily see their presence or lack thereof.



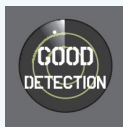
Also place a single Fighter unit of each type contained within the detected force.

Example: *If the Task Force contains 2 US carriers and the French carrier 'Foch', place one F-14 and the F-8 Crusader Fighter unit on top of the Task Force.*

Design note: *Poor detection represents a large area in which the enemy could be operating or outdated information such as a 4-hour old detection report, but the presence of fighters will be revealed by radar emissions and communications intercepts).*

Good detection Task Forces reveal the entire force for inspection by the enemy.

Place the entire contents of the Task Force close to the Task Force marker for inspection by the enemy player. **Design note:** *A good detection represents an hour-old detection or very accurate location of the Task Force.*



11.2 Detection reduction

When ships move due to the 'Ships' event but not the 'Fast' event, the detection status of a Task Force deteriorates by one step from 'Good' to 'Poor' to 'Undetected'. Do not remove Convoys or ships already placed close to Task Forces. These are only removed if the owner conducts an action (using OPS) which allows moving counters from one Task Force to another [16.4].

11.3 Casualties Against Hidden Task Forces

Occasionally units within a Task Force are hit, and the attacker is allowed to select a unit to take the damage, but the Task Force is hidden from view (because it is only poorly detected). When this happens the owning player reveals the counters of units which could be hit for the enemy to make a decision. This is temporary and does not increase the Detection level of a Task Force.

12.0 TASK FORCE MOVEMENT

Task Forces may move one zone in any direction desired. All 'Good Detected' Task Forces move first, followed by all 'Poor Detected' Task Forces and finally all 'Undetected' Task Forces. The Soviets must move all Task Forces first during each category (Good-Poor-Undetected) followed by NATO.

12.1 Bad Weather Effects on Task Forces

Any zone currently experiencing Bad Weather slows down Task Force movement. Instead of moving, place a 'Moved' marker on a Task Force in Bad Weather. On the next opportunity to move remove the marker and move normally. Repeat for each Task Force starting in a zone with Bad Weather.



12.2 Amphibious Landing

Instead of moving, a 'Landing' marker may be placed on a Task Force in a zone with an Amphibious Landing symbol. It does not move from the zone but may instead perform an Amphibious Landing on its next chance to move. Roll a free Task Force Detection roll (17.10) against each TF marking itself as 'landing'.

In **Bad Weather**, first place a 'Moved' marker and then on the next chance to move you may choose to move normally or place a 'Landing' marker.

Both sides have Amphibious units which represent landing ships loaded with troops and equipment, these must be within a landing Task Force to make an Amphibious Landing.



12.3 Entering or Leaving a Port

Instead of moving from a zone, when allowed to move a Task Force some or all of its units may move into a port adjacent to the Sea Zone it is currently in. When leaving a port, ignore Bad Weather. Reload units entering a port as appropriate [13.9].

12.4 Dissolving a NATO Task Force

A NATO Task Force delivering a Convoy to a Convoy destination port or located in the Bay of Biscay or located in the Mediterranean 1-2 Sea Zone may choose to dissolve itself to free up a Task Force marker. This also allows the units to return to the US without the NATO player having to manage this process.

If the NATO player wishes to dissolve a Task Force, place the entire contents of the Task Force on the Turn Track underneath a 'Dissolved Task Force' marker **two** turns from the current turn or **three** turns from the current turn if the Task Force was in the Mediterranean. They re-enter the game as reinforcements, all arriving in the USA regardless of nationality. In addition, some or all of these units may join one or more existing Task Forces within two zones of the USA when they re-enter the game. If entering in the US, remove all ammunition markers from the units.

12.5 Task Force Movement Protocol

To aid play of the game, when a Task Force moves, place it into the new Sea Zone close to the border of the old zone – in this way your opponent can clearly see which zone it moved from.

13.0 SPECIAL UNITS & CONCEPTS

13.1 Soviet Tanker units

The Soviet player has Tanker units which can extend the range of land-based Strike aircraft. Each 2-step Tanker unit committed to a Strike mission may extend the range of one Strike unit by 2 zones or two Strike units by 1 zone. A one step Tanker may extend the range of one unit by 1 zone. Select the tanking zone by placing the Tanker unit the number of zones from its base as specified above; the Strike mission then counts its actual used distance from this zone.

If the tanking zone contains enemy Interceptors, resolve Interception detection/attacks as normal. Loss of a Tanker unit has immediate effect in terms of strike range.

Example: A Soviet strike departs the Kola Peninsula with two Badgers. One Tanker unit allows two Strike units one extra point of range, so their transit to the Barents Sea is 'paid for' by the tanker. They refuel over the Barents Sea and continue to East 3, which is 3 zones away at their maximum range.

13.2 Convoys

Convoy counters represent 40-50 merchant ships laden with supplies.

A Convoy counter's worth degrades based on the number of hits taken to reach the Convoy destination port. Some of the Convoys have intrinsic defenses printed on the counter showing SAMs and the Anti-Submarine Warfare value available to the Convoy.

When entering or leaving a port, roll for Mines and evaluate any Port Damage. (See charts below) When a Convoy arrives in a Convoy destination port it is evaluated to determine its worth.

Mines	Effects
Light	2 dice
Heavy	4 dice
Each '1-2' causes a hit with all odd hits being allocated to a Convoy by the Soviet player	

Port Damage	Effects
Light	+1 hit to each Convoy
Heavy	+2 hits to each Convoy

The extra hits caused by port damage represent the convoy being more time consuming to unload due to damaged port infrastructure, etc and so the convoy ultimately takes an extra day or two to unload.

Hits	Convoy Effects
0-2	NATO Supplies Marker
3-4	NATO Partial Supply Marker
5-6	No effect
7-9	Convoy Massacre
10	Additional Convoy Massacre and remove from the map

The delivery of a Convoy to any Belgian port, to the Italian port of Taranto and to any Turkish port allows the Convoy to still have some value despite the following damage:

Delivery to	Effects
Belgian Port	Treat 5 hits as a Partial Supply
Taranto	Treat 5 hits as a Partial Supply
Turkish Port	Treat 5 and 6 hits as a Partial Supply

13.3 Convoy Massacres

When the Soviets score a total of seven hits on a single Convoy, and again when the total reaches ten hits on the same Convoy give the Soviet player one Convoy Massacre marker.



When the Soviet player has two Convoy Massacre markers in-hand, add two Operations Points to the current Soviet total. When the Soviet player has three Convoy Massacre markers in-hand, add one to the NATO losses score, then return all Convoy Massacre markers to the pool.

Note: A Convoy which takes a tenth hit is removed from the map.

Design note: Ten loss points represents the loss or heavy damage of about forty merchantmen of a total of approximately eight hundred available to NATO – or ~5% – and that total would have been at full mobilization. A more realistic figure is five to six hundred ships for all but the 'Buildup' scenario.

13.4 Amphibious Landing Units

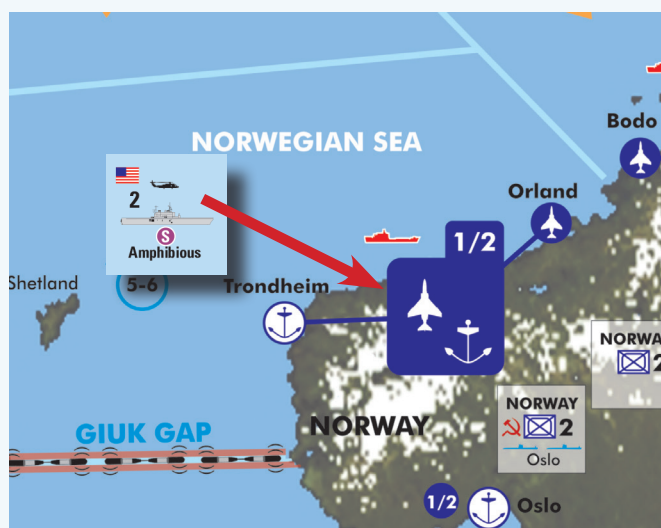
Both sides have Amphibious units which represent landing ships loaded with troops and equipment.

When they are in an appropriate Sea Zone, they may conduct an Amphibious Landing.

Possible Soviet landing sites are marked on the map.



NATO can land troops only in friendly ports which does not use the Amphibious Landing rules. Exception: Soviet occupied Iceland [21.4]



13.5 Follow-on Soviet supply groups

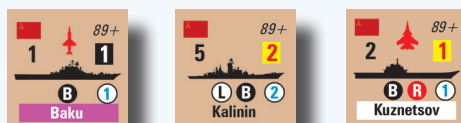
As well as full-strength Amphibious units, the Soviet player receives additional units during the game, listed on the Turn Track. They are treated the same as full-strength units except that they start with two hits and are therefore worth less Troops markers when landing.

1 Day 1 & 2	2 NATO Med Convoy Day 3 & 4	3 NATO Med Convoy First Strike 6+ Day 5 & 6	4 Soviet Supply Day 7 & 8	5 NATO Med Convoy First Strike 6+ Day 9 & 10	6 Soviet Supply Day 11 & 12	7 NATO Med Convoy First Strike 6+ Day 13 & 14	8 Soviet Supply Day 15 & 16	9 First Strike 6+ Day 17 & 18	10 Last NATO Convoy Day 19 & 20	11 Day 21 & 22	12 Day 23 & 24
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MISSILES		1	2	3 Next Day	4	5 Ships	6 Repair	7	8 Fast	9	Day One
1	11	10	11	12	13	14	15	16	17	18	Day Two No Fast
2	12										

13.6 Soviet Admiral Kuznetsov Aircraft Carrier

The unit is included for the 1989 scenario as it is assumed that the USSR wasn't falling apart and managed to finish construction and commissioning. You may select any two of the three additional capital ship units made available if playing in 1989.



Pick 2 of the 3

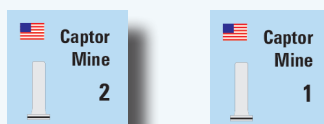
13.7 CAPTOR Mines

These mines are effective only against submarines, they are placed through card play.

When an enemy submarine moves into or through a Captor mined zone roll two dice (one die if the CAPTOR Mine counter is on its step loss side) using the Anti-Submarine Warfare process.

Bad Weather removes one die (to a minimum of 0) but there are no other possible modifiers. If a one is rolled, flip the Captor Mine counter or remove it if already on its step loss side (Soviets are mapping the field) and otherwise ignore those dice.

A submarine which is part of a Task Force is ignored for the purpose of this rule.



13.8 Port Mining

Ports can be mined via card play.

Each submarine entering or leaving a mined port must roll one die and take a step loss on the roll of '1-2'. Roll one die per Submarine unit for a 'Lightly Mined' port or two dice per unit for a 'Heavily Mined' port. Roll one less die if the submarine is small (S) or non-magnetic (N) to a minimum of no dice.

Task Forces roll two dice ('Light') or four dice ('Heavy') with each roll of '1-2' causing one hit. Odd hits must be applied to a Convoy or Amphib unit if present. Other hits are allocated freely to the Task Force by the player taking the damage.



13.9 Missile Ammunition Tracking

Only units with an (L) designation track SAM ammunition usage. They never have to fire, only if the owner wants them to. When they do, they receive the Ammunition marker for that unit as specified below.

13.10 Ticonderoga VLS (Vertical Launch System)

If it fires between 2 and 4 SAMs, mark it '2-4 Fired' - it now has a SAM value of 4.

If it repeats this, mark it Empty - it now has a SAM value of 0.

If it fires 5 or more in one salvo, mark it Empty.

It can be reloaded only in a Major Port.



13.11 Kirov and Kalinin - Soviet named Capital ships

If they fire 2 or more SAMs, mark '2+ Fired, One Volley Left'

If it repeats this, flip the ammo marker over- it now has a SAM value of 1 Kirov, 2 Kalinin.

They can be reloaded only in a Major Port.



Design note: These units have many point defense missiles, with a huge quantity of ammo.

13.12 Cruise Missile Ammunition

All units with cruise missiles receive a 'Cruise Missiles Fired' marker when they have fired them. They cannot be partially used. When fired, they are all fired.

For the Soviets, use this marker for green, yellow and black missiles, and white missiles from submarines (anything capable of also attacking land targets)



13.13 Anti-Ship Missile Ammunition

All units with anti-ship missiles receive an 'Anti-Ship Missiles Fired' marker when they have fired them. They cannot be partially used. When fired, they are all fired.

For the Soviets, use this marker for red and white missiles, except white missiles from submarines (which are short range and anti-ship only)

Design note: This system allows for an easy way to track ammunition without complicated bookkeeping.



13.14 Soviet RORSATs

The Soviet player has reconnaissance satellites (RORSAT) for the detection of Task Forces.

At the start of the game most are on the ground at Tyuratam but they can be launched for 2 OPS each. When launched a die roll of '1' results in an unsuccessful launch and the satellite is instead destroyed - place in the Destroyed box.

For two OPS the Soviet player may also take a RORSAT on the map and move it anywhere. Again, a die must be rolled and the RORSAT is destroyed on a '1'.



RORSATs can only be newly stacked together in the same zone if there are at least four already in play (in space). For clarity, the fourth one launched can be stacked together with another. Once stacked together they roll more dice for detection (1 die per RORSAT present in the zone).

RORSATs can be removed from play due to collateral damage caused to Soviet facilities [17.22]. If the **Lourdes Facility** is destroyed [21.5], remove one RORSAT from the game permanently (selected by the US player on a die roll of 6+, otherwise by the Soviet player) and remove another temporarily returning three turns from the current turn (selected by the US player on a die roll of 6+, otherwise by the Soviet player)

Design note: The loss of data downlink and analysis capability would have hindered Soviet targeting.

13.15 Continental United States

F-15 Fighter units are printed on the map in the US zones. The NATO player may use an unlimited number of F-15's for CAP when defending in these zones.

14.0 CAPITAL SHIPS

All aircraft carriers and any ship with limited ammunition (L) or a damage code (B) or (H) is a capital ship, except for the US Amphibs - which are amphibious units.

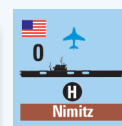
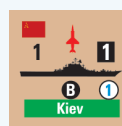
Capital ships are harder to kill, because they are in the centre of the formation and heavily defended.

Capital ships with a damage modifier symbol (B) and (H) roll to determine the effect of any hit.

Those without a value take a step loss normally. **Design note:** These units represent multiple ships.

When initially damaged, half (round up) of all Air Unit steps based on the carrier must also roll to take a step loss on a 6 +. The owner selects those to roll for and applies step losses received.

If a carriers sinks, the Air Units onboard are destroyed and placed in the Destroyed box.



Carriers without a damage modifier automatically lose half (round up) of their Carrier Air Group per step loss taken and all of the Carrier Air Group if they sink but are never damaged because the counter represents multiple carriers.

14.1 Capital ship Damage

To determine the effect of a hit on a capital ship with a damage modifier (B) or (H), calculate the number of dice to roll from the table below, and then apply the highest single die on the result table:

Damage Dice to Roll	
1	NATO missile, Soviet white missile from a ship Soviet red missile, Tomahawk missile (card).
2	Soviet yellow, black or submarine white missile, Soviet or NATO torpedo
3	Soviet torpedo from (T) submarine
+1	Already damaged
+1	Each additional hit taken at the same time

Die Roll	(H) Huge	(B) Big
Sunk	Any 10	Any 8, 9 or 10
Damaged (flip)	Otherwise	Otherwise

Damaged carriers cannot fly Strike aircraft. CAPs receive a modifier when resolving Fighter vs Air Strike combat and a carrier no longer counts for Submarine vs Task Force attacks (it has less helicopters in the air). It also loses its intrinsic detection capability.

14.2 Capital ship Break-off

A damaged capital ship may immediately, and for free, form itself and any escorts from its previous force into a new Task Force with the same detection status as the previous one. This allows the damaged ship to break away from the Task Force and return to a base.

14.3 Capital ship Intrinsic Maritime Patrol

Each undamaged aircraft carrier (showing an Aircraft symbol) or helicopter carriers (showing a Helicopter symbol) has one **intrinsic Maritime Patrol unit** (representing AEW, helos etc.) it is usable only for detecting enemy Task Forces within the carrier's current zone. Each may roll one die on the Task Force Detection Table [17.10] during the 'ships move' event.

Exception: US carriers roll two dice each.

Exception: US Amphibs do not have an intrinsic detection capability

When intrinsic MP are subject to Interception or CAP detection, when rolling for Fighters vs MP attacks treat US carrier intrinsic MP as planes and all other carriers as MP helicopters.

Example: *A British Task Force and a Soviet Task Force are in the same Sea Zone. British Harriers on board Illustrious will be able to roll to 'bounce' Kiev's intrinsic MP unit before it can roll to detect the British Task Force. This will be resolved as a Fighter vs MP attack against a MP helicopter – because Kiev is operating helicopters in the AEW role.*

15.0 REPAIRING CAPITAL SHIPS

A capital ship which is damaged but arrives at a Major Port must roll immediately for repairs.

Place the unit the indicated number of turns in the future on the Turn Track at full-strength.

If the unit is a carrier, place its air complement with it on the Turn Track. A damaged capital ship takes no part in the game until it returns to play

Die Roll	Damaged Unit
1-3	5 turns
4-7	4 turns
8-10	3 turns

16.0 OPERATIONS POINTS

When a card is played for Operations Points it generates the number of Operations Points referred to as 'OPS' indicated in the top left corner. This will occur twice per turn for each player. All other cards played will be Reaction Events, using the bottom event on the card and costing no Operations Points to play - discard the card after resolving the event. (see 16.6.5 for an additional way to play the top Operations Event on a card in your hand).

16.1 Actions

When it is your turn to perform an action (the Soviet player has the first play each turn) you may select any action to resolve from the Actions Chart or pass your turn. Each action is described below.

Note: When a player passes, they must have less OPS than their opponent, the other player takes an action and the passing player then has another chance to play.

16.2 Submarine units

The following rules apply to all Submarine actions:

Unless specified, all actions are performed with a single submarine and cost 1 Operations Point.

All units taking an action must be fresh.

Diesel submarines (D) move 1 zone only and cannot perform a 'Fast' move.

Small and Slow (S) diesel-powered submarines can move OR attack OR go on patrol. If moving, the player may move any two (S) submarines instead of the usual one.

Nuclear submarines move two zones, except with a 'Fast' move which allows 3 zones of movement.

A submarine attack can be either an anti-submarine or Task Force attack, or if missile armed a missile attack against a Task Force or a facility.

A submarine which takes an action will remain 'Spent' or 'On Patrol' for the rest of the turn.

A moving Soviet SSBN may take a non-diesel submarine with it for no extra cost, both are marked as 'Spent'.

Submarines can leave a Task Force they are with to perform one of the actions listed above. Submarines join a Task Force when ships move – see the 'Ships' event [9.1.3].

Design note: A submarine can only detect an enemy Task Force via an 'Anti-Submarine Warfare' attack and requires a natural 10 to do so [17.8]. This means submarines are poor at detecting Task Forces!

Applicable Submarine actions:

- 1 submarine moves and then attacks.
- 1 submarine attacks and then moves.
- 1 submarine 'Fast' moves (not diesels (D)) 3 zones.
- 1 submarine moves and goes 'On Patrol'.
- Missile coordination: 2-3 subs all move to the same zone and then resolve a missile attack.

16.2.1 'On Patrol'

A submarine may move and place an 'On Patrol' marker instead of attacking.

This allows one 'On Patrol' unit in a zone to interrupt a moving enemy submarine later in the turn during their action and roll an Anti-Submarine Warfare attack against it or to roll to attack a moving Task Force entering the zone, then flip the 'On Patrol' marker to 'Spent'.

Units 'On Patrol' can remain so indefinitely.

A unit which is 'On Patrol' cannot have an action spent on it and can only react to enemy movement into or through their zone.

16.2.2 Missile Coordination

Two or three submarines can perform this action together. All may move to the same zone and perform a missile attack against a single Task Force [17.12] or a facility [17.19] in range.

This action costs one Operations Point per submarine taking part.

16.3 Air Units

The following rules apply to all Air Unit actions:

All actions cost one Operations Point per flying unit except for escorting fighters and supporting Tankers which are free to fly with any air strike mission.

All units taking an action must be Fresh.

All Air Units must always remain within range of their base.

An Air Unit which takes an action will remain 'Spent' or 'On Patrol' until ships move or the end of the turn. Exception: Maritime Patrol units only remove 'Spent' markers at the end of the turn.

Design note: This means that other than Maritime Patrol Air Units could fly up to three times in a turn.

Applicable Air Unit actions:

- 1 Maritime Patrol unit fly and attack (Anti-Submarine Warfare) or detect Task Forces

Note: If a detection is placed or upgraded on an enemy Task Force, the active player may take one further action of any type as long as the action results in an attempted attack against one of the Task Forces detected.

- 1 MP/fighter fly and go 'On Patrol'
- 1 or more Strike units fly and attack - Fighters and Tankers fly for free with air strike missions.

16.3.1 'On Patrol' with Maritime Patrol units

A Maritime Patrol unit may go 'On Patrol' instead of attacking – move the unit to a zone within its range, this allows a Maritime Patrol Air Unit to interrupt a moving enemy submarine later in the turn during its action and roll an Anti-Submarine Warfare attack against it or roll to detect a moving Task Force entering or moving through the zone, then flip the 'On Patrol' marker to "Spent" & immediately return the MP unit to its base. Enemy fighters present may intercept and resolve CAP before the MP rolls.

A unit which is 'On Patrol' cannot have an action spent on it and can only react to enemy movement.

Air Units that are 'On Patrol' can remain so indefinitely or the marker can be removed, and the unit placed back at its base 'Fresh' when it is your turn to perform an action.

Note: It is not required to use an 'On Patrol' marker since an MP unit at sea must be On Patrol.

Additionally, all Task Forces in a Sea Zone with friendly 'On Patrol' Maritime Patrol units benefit from an additional defensive bonus when attacked by submarines (this is built into the combat table).

16.3.2 'On Patrol' with Fighter units

A land-based Fighter unit may go 'On Patrol' in a Sea Zone.

For Fighter units this is called 'Interception' – move the unit to a zone within its range.

Note: Carrier based Air Units cannot perform this mission type.

This allows a Fighter unit to attack any Air Units entering its zone and does not spend the unit.

Units that are On Patrol can remain so indefinitely, or the marker can be removed, and the unit placed back at its base 'Fresh' when it is your turn to perform an action.

Note: Sending a Fighter unit on an Interception mission into a Sea Zone already occupied by an enemy fighter or carrier-based CAP will trigger combat [17.2 and 17.3].

Design note: It is not required to use an 'On Patrol' marker since a Fighter unit at sea must be intercepting. Carrier Air Units cannot fly this mission type as they would be required to protect their carrier at all times. There is also a 'free' interception allowed if an enemy strike enters and then exits a zone containing a carrier (even an undetected one) [17.2].

16.3.3 Strike units fly and attack

One or more Strike units and any number of Fighters and Tankers may fly to a target Task Force or facility and attack.

Strikes by units from different bases are always resolved as separate strikes (carriers at sea in the **same zone** may fly as one strike). Strikes against different targets are always separate strikes – a strike versus a port/airfield in the same target area is the same target and is therefore one strike.

Note: Fighters may fly part way. If they fight at all, the whole strike is automatically detected thereafter. This can be useful for escorting a strike part way over Denmark or Turkey.

16.4 Surface Units

The following rules apply to all Surface Ship unit actions:

Ships may never move unless they are part of a Task Force [11.0] when not part of a Task Force, a Surface Unit must be in a friendly port facility. Surface Units are never marked 'Spent'.

Applicable ship unit actions:

• Form a Task Force (at a port or at sea where there is an existing Task Force).	Cost 2 OPS
• Move ships between Task Forces in the same zone.	Cost 1 OPS
• Attack with missiles from a Task Force.	Cost 1 OPS

16.4.1 Form a Task Force

Create a Task Force in a port with any number of ship counters and up to one nuclear submarine.

The British and northern French ports may be counted as the same port when forming a Task Force. Choose an available Task Force marker from the Task Force display and place the ship counters into the holding box. Forming a Task Force costs 2 Operations Points.

A Task Force may also be created at sea. To do so you must already have a Task Force at sea in the zone. Move any units from other Task Force(s) in the zone to this newly created Task Force. Any unit moved from a detected force grants that detection status to its new Task Force – place an additional Detection marker on the new Task Force. If there is a friendly nuclear submarine in the Sea Zone which is not Spent, it may join the new Task Force.

This action may also add ships from a port to an existing Task Force in an adjacent sea. The TF loses its next chance to move.

16.4.2 Move ships between Task Forces in the same zone

Move any units from one Task Force to another existing Task Force(s) as desired.

Any unit moved from a detected force grants the detection status of the originating Task Force to the new Task Force unless the new Task Force already has a higher detection status. This action costs one Operations Point to perform. If there is a friendly nuclear submarine in the Sea Zone which is not Spent, it may join any eligible Task Force.

Example: *TF Aegis and TF Hammer are in the same Sea Zone. TF Aegis is marked 'Poor' detected and wishes to send an escort surface unit to TF Hammer which was recently attacked. It does so, and TF Hammer is marked with a 'Poor' detection – the Soviets tracked the escorts as they transferred and now both Task Forces are marked with a 'Poor' detection marker. If TF Hammer was already marked with a 'Good' detection, no detection status would be altered.*

16.4.3 Attack with missiles from a Task Force

A Task Force may attack eligible targets with missiles [17.12]. This action costs one Operations Point to perform. White, blue and red ship-launched missiles do not use this action and can be launched only via card play.

16.5 Miscellaneous Actions

• Repair a Facility	Cost 2 OPS
• De-mine a Port	Cost 2 OPS
• Launch or move a RORSAT	Cost 2 OPS
• NATO may attack Cuba	Cost 6 OPS
• Pass, if you have less OPs than the other player	

16.5.1 Repair a facility

Remove a Hit from a damaged facility. If the removal would alter the damage status (from 'Heavy' to 'Light' or from 'Light' to not damaged), a die roll of 5 or higher is required to alter the status, otherwise the action is wasted. See [6.10.4] for the effect of damage on facilities.

16.5.2 De-Mine a port

Roll a die requiring 5+ to reduce a minefield by 1 level, or 7+ if the facility is damaged.

'Heavy mining' reduces to 'Light mining' which reduces to unmined and removes the Mine marker.

16.5.3 Launch or move a RORSAT

Launch a RORSAT from Tyuratam or move a RORSAT already on the map to another zone.

This action costs 2 Operations Points and fails catastrophically on a die roll of '1' which destroys the satellite instead. See [13.14] for full RORSAT rules.

16.5.4 NATO may attack Cuba

See [21.5] for details.

16.5.5 Pass

If you have less Operations Points than your opponent, you may pass and do nothing. Your opponent will then play his action after which you will have another chance to act.

16.6 'Use when active – not an action'

These can be used when it is your turn to play an action but do not count as an action or cost Operations Points to use.

- Use a First Strike Point to buy an effect (Soviet player only).
- Use a Soviet spy (Soviet player only).
- Play an event on either of your cards.
- Play a card in your hand's top (non-Reaction) event, adding two additional points to the listed cost.

16.6.1 Use a First Strike Point to buy an effect (Soviet player only)

The Soviet player may spend First Strike Points on several different options. See [22.1].

16.6.2 Use a Soviet spy (Soviet player only)

The Soviet player may use Soviet spies on several different options. See [24.0].

The use of a spy may reduce the Soviet Spy Infiltration Track, representing the discovery of the spy network.

16.6.3 Play an event on either of your cards

During the turn each player will use two cards to generate Operations Points. Each card has one event which can be used for the cost in Operations Points printed on the right hand side of the card. Each card can be used only once unless specified on the event text. Either card event can be used whenever allowed – they do not 'expire' until the end of the turn. When used, place a "Used" marker on the card.

16.6.4 Return an 'On Patrol' Air Unit to base

Return 'Fresh' 'On Patrol' Air Units per 16.3.1 and 16.3.2.

16.6.5 Play a card in your hand's top (non-Reaction) event, paying two additional OPS

You may play the top (non-Reaction) event of a card in your hand, paying an additional two points, plus the cost of the event itself. Place the card in the discard pile or remove it from play if the event instructs you to do so.

Design note: This allows you to keep events hidden in your hand to use at a time of your choosing.

16.7 'Use at any time (not an action)'

These can be used when you or your opponent are resolving an action.

- Play a Reaction Event when it applies.
- On Patrol units may roll to attack enemy units moving through their zone
- CAP may roll to attack Air Units attacking their base
- Hydrophone detection when enemy submarines move across the Hydrophone Barrier.
- Draw a card if you have less than three cards in your hand

16.7.1 Play a Reaction Event when it applies

Play a card in your hand as a reaction to something occurring. The card will specify when it is playable. Resolve it and discard it. If you drop below three cards in your hand, draw up to three cards.

16.7.2 On Patrol units may roll to attack enemy units moving through their zone

As per 16.2.1, 16.3.1 and 16.3.2.

16.7.3 CAP may roll to attack Air Units attacking their base

See [17.2] for resolution of a CAP Interception.

Note: All fighters are always performing a CAP mission and protecting their base, even when 'Spent' (except for 'On Patrol' interceptors, who are flying an Interception mission).

16.7.4 Hydrophone detection when enemy submarines move across a Hydrophone Barrier.

See [6.8] for detecting enemy submarines with hydrophones.

16.7.5 Use a Soviet spy (Soviet player only) for SSBN evasion

The Soviet player may use Soviet spies for SSBN evasion during the NATO action. See [24.0] The use of a spy may reduce the Soviet Spy Infiltration Track, representing the discovery of the spy network.

17.0 COMBAT SYSTEM

There are several types of attack/search actions in the game. The unit types and the attacks they can perform are listed below. To resolve an attack, select an attack type available to the unit and go to that section to resolve it.

Submarine	Anti-Submarine Warfare, Submarine attacks on Task Forces, Missile attack
Maritime Patrol	Anti-Submarine Warfare, Task Force detection
Air Strike	Missile attack, Bombing
RORSAT	Task Force detection
Task Force	Anti-Submarine Warfare, Missile attack
Carrier	Task Force detection (by Intrinsic Maritime Patrol)
Fighter	Escort an Air Strike, CAP, Interception, Fighter vs Maritime Patrol

How to Resolve an Air Mission

17.1 Transit to a zone

Air Units must trace their path to a Sea Zone to show how they arrived there.

- 1) Check the route TO (but not back from) the target for Interceptions and resolve each in turn.
- 2) Resolve any Interception in the target zone.
- 3) Resolve CAP by eligible units protecting the attacked target.

Example 1: *A Bear-D flies to a zone of interest. A path is traced by the Soviet player and Interceptions are resolved as it flies to the zone. When it arrives in the zone it may be first intercepted and then attacked by any CAP. Only after this can the Bear perform its mission.*

Example 2: *The Soviet player has a 'Good Detection' on a NATO Task Force and wants to attack it. A strike is flown, and the Soviet player indicates its route, resolves all Interceptions along the way and then any CAP within the target zone and finally attacks the target.*

17.2 Interception and CAP

Air Units are intercepted when:

R#2

- Present within a zone with an enemy air unit on an Interception mission. (On Patrol)
- Leaving a zone with an enemy aircraft carrier (you may not know it's there so the choice of whether to declare the presence of one or more carriers and roll the dice is up to your opponent).
- Overflying Denmark or Turkey. The War Track Fighter symbols show the number of dice to roll. [20.0] This number of dice is rolled once against the Strike mission, not once per unit flying. Use the values of the Fighter symbols printed on the map

17.3 CAP may roll to attack when:

- A strike attacks a location being protected:

Carrier based fighters defend every Task Force in the Sea Zone and every land facility adjacent to the zone (the carrier does not have to be a target).

Land based fighters protect land bases in their zone (but not Task Forces out at sea).

R#2

- An Air Unit is present within a Sea Zone with carrier-based FTR prior to completing whatever mission it was sent to perform.

Example 1: *MP aircraft are attacked by CAP prior to attempting Task Force detection.*

Example 2: *A fighter is given an Interception mission in a Sea Zone which is occupied by a carrier. The carrier CAP may attack the interceptor.*

17.4 Types of air to air combat

When Interception or CAP attacks occur they are resolved depending on what the triggering unit is:

If the unit triggering the Interception or CAP response is a MP unit, use 'Fighters vs Maritime Patrol'

If the unit triggering is an air Strike unit, use 'Fighters vs Air Strike' or 'Fighters vs Escorted Air Strike' if the strike has escorting fighters with it.

If the unit triggering is a Fighter unit use 'Fighter vs Fighter'

17.5 Fighter vs Maritime Patrol

One 'Fresh' Fighter unit may roll its printed air to air dice minus two dice (minimum one die).

A "Spent" fighter may not roll at all. Add the fighter's Tactical Value to each die rolled. Add +1 DRM if the Soviets have the 'Air to Air Missile' Technology marker.

For each result of 9 or more (6 or more vs a helicopter MP unit) the MP unit is disrupted.

On a natural 10 (N8-10 vs a helicopter MP unit) the unit takes a step loss and returns to base 'Spent'.

A disruption result returns the MP unit to its base "Spent" if it was conducting an Anti-Submarine Warfare mission, or going on-patrol. If it was resolving Task Force detection, it will roll one die less which may reduce it to zero dice – in which case it returns to base 'Spent'.

R#14

Fighter Combat vs Maritime Patrol (MP)	
Includes vs Intrinsic carrier MP	
Roll air-to-air dice -2 (min=1) with one fresh fighter	
Die roll + Tactical Value = kills allowed	
9+ vs plane	- 1 search, per die
N10 vs plane	Step loss and no search
6+ vs helo	-1 search, per die
N8-10 vs helo	Step loss and no search
Air to Air Technology- Soviet +1 DRM	
Intrinsic MPs are always helos except for US carriers	
N: means a natural roll	

17.6 Fighter vs Fighter

If both sides have fighters in the same Sea Zone the Soviet player must declare if he will avoid combat followed by the NATO player. If either does so, return interceptors to base 'Spent', and place a 'Spent' marker on CAP units. They take no further part in this combat. Otherwise, the opposing fighters will attack one another as follows:

At most one fighter may roll for each enemy fighter present - roll their air to air dice, applying modifiers and add their Tactical Value to each roll. (minimum one die per unit)

Fighter vs Fighter Modifiers	
Intercepting unit	- 1 dice
CAP "Spent"	-2 dice
Detected by Interceptors	+1 dice
Damaged carrier OR base light dmg	-1 dice
Soviet 'Air to Air Missile Technology' marker	+1 DRM
Base heavy damage	= 1 Die

R#4

Each die roll that is equal or higher than an enemy fighter Defense value causes a step loss.

All dice rolled are allocated by the rolling player as desired.

All step losses caused are simultaneous.

17.7 Fighter vs Air Strike

Interception and CAP is resolved against air Strike units using the following process:

One Fighter unit may roll its printed air to air dice for each enemy Strike unit in the zone, but a given Fighter unit may only roll its printed air to air dice once.

For CAP, the defending TF must use its own fighters first, if able, and if its own fighters are no worse than other fighters which could

be chosen (based on highest relevant values – air to air and tactical).
Modifiers are applied as below – apply them per unit rolling.
(minimum one die per unit).

Fighter vs Air Strike Modifiers	
Intercepting unit	- 1 dice
CAP “Spent”	-2 dice
Strike detected by Interceptors	+1 dice
Base light damage or damaged carrier	-1 dice
Base heavy damage	=1 dice
Soviet Air to air Technology marker	+1 DRM

For each die roll, add the rolling fighters’ Tactical Value.

Each total equal to or higher than an attacking Strike unit’s Defense value causes a step loss. The rolling player may allocate all rolled dice to targets, and then the player being shot at can make saving throws based on the detection status of the Task Force/base from which the CAP originated.

Detection Status and Air Strike Saves	
Good detection on shooting carrier CAP or CAP from any airfield	Attackers gain 6+ save (vs. CAP only)
Poor detection on shooting carrier CAP but only vs. Task Forces not being attacked	Attackers gain 8+ save (vs. CAP only)

Example: *Two Fighter units on CAP may roll their dice because there are two attacking Strike units. Each gains +1 die if any Strike unit was detected by an interceptor on the way to (or within) the target zone. One is from a Task Force with a ‘Poor’ detection on it which is not being attacked – any hits it causes can be ignored on a saving throw of 8 or more. The other is from the target Task Force, which has a Good Detection marker on it – any hits it caused can be saved on a roll of 6 or higher.*

17.7.1 Killing air Strike units before weapon launch

A natural ten causes a CAP to shoot down an enemy Strike step before the Strike unit launches its missiles – they are not included in the attack.

A natural 9 or 10 causes a CAP to shoot down an enemy Strike step using bombs before the Strike unit drops the bombs – they are not included in the attack.

Interceptors always cause step losses before strike units attack.

Note: Because fighters may kill attacking steps before they fire their missiles and drop their bombs, it’s a good idea to use the ‘Missiles’ marker to record the number of missiles launched or the ‘Steps Bombing’ marker to record the same before any CAP dice are rolled.

17.7.2 F-14 anti-missile capability

Each natural one or two rolled by an F-14 fighter shoots down that number of missiles launched by air Strike units when attacking a Task Force.

Design note: *F-14 Phoenix missiles had an anti-missile capability for use against Soviet missile-carrying bombers.*

17.7.3 Interception Detection and Warning

Interceptors which successfully kill any step of enemy aircraft, fail to kill by one pip or fight other fighter aircraft cause the air strike (or en-route Interception mission) to be considered detected for the remainder of the resolution – therefore all further fighters which engage the strike gain +1 dice.

If an air strike has been intercepted, after all fighters have made their rolls in that Sea Zone the owner may voluntarily abort the strike – return it to base ‘Spent’.

17.7.4 Fighter vs. Escorted Air Strike

Resolve exactly as for Fighter vs Fighter combat, but the non-striking player may roll with a number of fighters equal to the number of enemy Strike or Fighter units. All damage must be allocated to Fighter units first except that a natural ten rolled by any defending fighter may be allocated to a Strike unit.

If all escorting FTR are destroyed, hits can then be applied to STK. A natural nine may also be allocated to a Strike unit if the strike unit is bombing. Any Strike Air Unit taking a step loss in combat with fighters on both sides do not receive a saving throw for detection status and all kills take place before weapons are launched.

17.8 Anti-Submarine Warfare

Usable by: Submarine and Maritime Patrol units, Task Forces choosing not to move and perform ASW.

Purpose: Destroy enemy submarines, detect Task Forces, roll on the NATO Boomer Hunt table.

- 1) Roll the unit’s Anti-Submarine Warfare dice and apply modifiers.

If a Task Force is resolving the activity, its component will determine the number of dice to roll. [17.9]

- 2) Add the Tactical Value of the rolling unit to each die. Each roll equal or above the Defense value of an enemy submarine in the same zone causes a step loss.
- 3) The defending player (the player not rolling) may allocate any one die which is not a 10 to any unit which could be destroyed with that die roll – thus a ‘9’ could be allocated to a much noisier ‘8’ boat, but an ‘8’ could not be allocated to a ‘9’ boat.

If the attacking unit is a Maritime Patrol unit or a Task Force, (N) submarines may only be killed by natural 10s.

Note: *Non-magnetic submarines are difficult for air units (and the helicopters from ships) to detect.*

- 4) The attacking player then allocates all remaining rolls as desired.
- 5) Natural ‘10’ rolls allow one of three possible outcomes– the rolling player may pick one outcome for each ten rolled.
 - Detect a Task Force in the same zone.
 - Destroy a step of any submarine selected by the attacker
 - The Soviet player may hunt NATO SSBNs [23.0].
- 6) A double which is not a 10 allows a currently rolling rocket-torpedo equipped submarine to cause a step loss to any target submarine in the same zone. **All Soviet submarines are rocket torpedo equipped.** NATO submarines do not have these weapons.

R#15

7) One or more natural 1's rolled when a NATO submarine is attacking, and that submarine is in the Bastion causes a step loss to the currently rolling Submarine unit.

8) Any 'save' submarine receives a 6 or higher save against any hit caused to it for any reason. Roll one die and ignore the hit if the result is 6 or higher.

Design note: Submarines can only detect Task Forces during an ASW attack and at no other time. Also, they require a natural ten to do so – submarines are not very good at detecting Task Forces.

Anti-Submarine Warfare Natural Rolls	
N 1	A rolling NATO submarine in the Bastion takes a step loss (one maximum).
Soviet- a double < 10	Rocket torpedoes – cause one step loss to any unit selected by the rolling player in the same zone. Ignore if in the Arctic.
N 10 Option #1	Detect an enemy Task Force Place a Poor Detection marker or improve a Poor detection to a Good detection on any enemy Task Force in the same Sea Zone. After spending one '10' the submarine may ignore all other dice (except for ones if NATO rolling in the Bastion) and may immediately attack the Task Force using missiles [17.12] or submarine attack on TF [17.11]
N 10 Option #2	Destroy a step of any submarine selected by the attacker .
N 10 Option #3	If a Soviet nuclear submarine is rolling and not in the Baltic, Barents, Norwegian (North 5-6 and 7-8), or Black Sea: Take the 'USSR SSBN Hunting' marker or discard one already held and Roll on the NATO Boomer Hunt Table [23.0].

Anti-Submarine Warfare Die Modifiers	
Hydrophones	+ 1 die
Bad Weather and non-MP unit detecting	- 1 die
Bad Weather and MP unit detecting	- 2 dice
Under Arctic ice	- 1 die
When under Arctic ice all submarines gain a 6+ save or increase a 6+ save to 4+. Rocket torpedoes do not function under the Arctic ice.	

Anti-Submarine Warfare DRMs	
Attackers Tactical Value (X)	+ X
Soviet torpedo technology and a nuclear (non-diesel) submarine is attacking	Tactical = 1

17.9 Anti-Submarine Warfare by Task Forces

A Task Force may roll dice using [17.8] Anti-Submarine Warfare based on its component ships instead of moving – calculate the TF ASW value and roll the dice listed with the Tactical Value. Ignore the fast modifier on the ASW total table.

If any ship has rocket torpedoes, then the TF may use that rule (through the rolling of doubles).

(N) submarines can only be killed on a natural 10.

R#16

Task Force ASW	
Unit	Value
Submarine	Anti-Submarine value x 3
Each Carrier	+2 ; 0 in Bad Weather
Each MP aircraft on patrol	Anti-Submarine value 1 in Bad Weather
Each Ship	Anti-Submarine value

Task Force ASW Total		
ASW Total	ASW Dice	Tactical Value
0	0	
1-2	2	
3-5	2	+1
6-8	3	
9-12	3	+1
13-17	4	
18+	4	+1
Move up one row in Bad Weather		
Move up one row if the Task Force is fast		

17.9.1 Penalty for ASW Operations

A Task Force which performs ASW is automatically marked with a 'Poor' Detection. In addition, if the Task Force rolls any two N1 or N2 results whilst rolling, instead upgrade to a 'Good' detection.

Design note: Both the Soviets and NATO experimented with surface ships hunting for submarines. The main problem with this is that submarines in general have better detection capabilities than ships and that to hunt effectively a Task Force would have to disperse making it an easier target to attack.

17.10 Task Force Detection

Usable by: Maritime Patrol units, RORSATs, Task Force Intrinsic Maritime Patrol.

Purpose: Detect enemy Task Forces so they can be attacked.

Units roll their 'Task Force Detection' dice after modifying the # dice to roll (which may be reduced to zero) through enemy fighter activity [17.5], on the following table. Resolve the dice from lowest to highest value rolled.

Die Roll	Task Force Detection
1-2	No Effect.
3-5	Poor detection. May upgrade any Poor to Good. Placed by TF owner unless upgrading.
6-8	Poor detection. May upgrade any Poor to Good. Placed by rolling player.
9-10	Good Detection marker placed by rolling player. Change the die to a 6 in Bad Weather, or to a 1 in Bad Weather if a RORSAT is rolling.

The result indicates which player places the detection - that player may place on any eligible Task Force. If a Task Force is already detected, the detection may be upgraded (from 'Poor' to 'Good') instead of placing a new Detection marker.

Task Force Detection Die Modifiers	
US Carrier intrinsic MP detecting	2 dice per US carrier
>2 Enemy TF in the zone	+ 1 die per TF over 2
Soviet ASAT system active (NATO only)	-1 DRM per FSP in the ASAT Box

Once a unit allocates all detection dice (not before) the Task Force owner must implement [11.1] and reveal the relevant information depending on detection status of the Task Forces. For ease of reference, this is summarized below:

Detection Fog of War Effects	
Poor Detection	Reveal Amphibs and Convoys present and one fighter of each type present.
Good Detection	Reveal entire group.

17.11 Submarine attacks on Task Forces

Usable by: Submarines

Purpose: Attack detected enemy Task Forces (with torpedoes)

Attack Process

1) The submarine player announces the target Task Force, which must be detected and then determines the number of dice the submarine will roll using the Anti-Surface value as the base value and then applying modifiers, to a minimum of one die.

Submarine vs Task Force Die Modifiers

Good detection on target TF	+1 die
Bad Weather	-1 die
Target TF is fast	-1 die. -2 if diesel submarine

2) The submarine player rolls the dice, adding the submarines Tactical Value to each and determines whether any hits have been scored. A submarine with (T) super heavy torpedoes may treat a 6 as a 7-9 and a 10+ as an N10 result, alter the dice now if this applies.

R#7

Submarine vs Task Force Attack	
Die	Result
1-5	Miss
6 (T)	Hit
7-9	Hit on PM / Amph / Convoy
10+ (T)	2 hits on Amph / Convoy (once only) or treat as 9
N10	Hit vs any or treat as 10+

R#7

3) If there are no hits, the combat ends now – the submarine failed to contact its target. Otherwise, leave the dice to one side for the moment and proceed to step 4.

4) Any dice which roll a value high enough to cause a step loss to an escorting submarine within the TF may be expended now to do so. The escort submarine receives saves as normal. Note that this happens before TF ASW is calculated

R#11

Then, If the submarine rolled a double (a pair) it may conduct an SSM (missile) attack if so armed – all US and UK nuclear submarines have these weapons as do Soviet nuclear submarines if Torpedo Technology is in effect.
-Cause a step loss as though rolling a 'Hit' result if the double is a 1-6
-Cause a step loss as though rolling a 'Hit on PM / Amph / Convoy' result if the double is a 7-10

The loss is allocated by the attacking player and takes effect immediately. Then discard the two dice used.

5) Calculate the Task Force ASW total using the following values:

Task Force ASW	
Unit	Value
Submarine	Anti-Submarine value x 3
Each Carrier	+2 ; 0 in Bad Weather
Each MP aircraft on patrol	Anti-Submarine value 1 in Bad Weather
Each Ship	Anti-Submarine value

6) Look-up the ASW total to determine the number of dice to be rolled by the Task Force owner and the value to be added to each die rolled (the Tactical Value)

Task Force ASW Total		
ASW Total	ASW Dice	Tactical Value
0	0	
1-2	2	
3-5	2	+1
6-8	3	
9-12	3	+1
13-17	4	
18+	4	+1
Move up one row in Bad Weather		
Move up one row if the Task Force is fast		

7) If the Task Force rolled a double (a pair) and any ship within the Task Force is rocket torpedo (R) armed and it is not Bad Weather, it may alter one of the dice to an 8 and discard the other die.

8) For each die, check to see if the roll plus Tactical Value is enough to cause a step loss on the attacking submarine (equal to or above its Defense Value) – if so, apply a step loss for each.

(N) submarines suffer a step loss on a natural 10 only

Note: Any die generated from step 7 does not benefit from a Tactical Value

Any 'save' submarine may roll normally to ignore a hit caused (6+)

9) For each roll of 8 or higher, the Task Force owner may select and remove one of the already rolled attacking submarines dice from those rolled in step 2, thus preventing the hit from taking place.

10) Evaluate the remaining submarine attack dice

Each '**Hit**' cause a hit to a non-Prairie Masker (P) equipped non-Capital Surface Unit which is not an Amphib or Convoy, selected by the attacking submarine. (i.e.: regular ships without PM).

Each '**PM / Amph / Convoy Hit**' cause a hit to any unit which is not a capital ship unless only capital ships remain within the Task Force. The attacking submarine may select the first of these hits scored, after which the Task Force owner may select the remainder unless the attacker chooses to apply all remaining hits to Convoys or Amphibious units.

A natural '10' result can cause a hit on any potential target, selected by the attacking submarine – this is how you sink capital ships.

17.11.1 Effect of Hits

Hits to ship units cause one step loss each.

Amphibious units and Convoys take one hit per hit scored.

Capital ship units follow the process in [14.1] per hit scored.

17.11.2 Illustrious UK carrier unit

These British small carriers ("through-deck cruisers") were fitted with Prairie Masker and receive a special save of 8+ against any torpedo hit. A successful roll ignores the hit entirely and causes the NATO player to have to sing 'Rule Britannia' for the save to count!

17.12 Missile attacks on Task Forces

Usable by: Missile-armed air, submarines and ships.

Purpose: Attack detected enemy Task Forces (with missiles from a distance)

Use the appropriate rule based on the attacking unit:

17.13 for submarine launched missiles

17.14 for surface ship launched missiles

17.15 for air launched missiles

then move on to the 'Resolving Missile Attacks' section [17.16].

17.13 Submarine Launched Missiles

The attacking submarine(s) must be in the same zone as the target Task Force.

Roll one die, applying modifiers and use the best die rolled to determine the result.

Submarine Launched Missile Modifiers	
Good detection on target TF	+1 die
Diesel attacker and target TF is fast	-2 DRM
Missile Coordination per unit after 1st	-2 DRM

Missile Attack Type	
1-3	No attack
4-7	Long Range
8-9	Short Range
10+	Short Range and Torpedo attack

No attack: No attack, mark 'Spent'. But no ammunition expended.

Long Range: Resolve a missile attack with the total number of missiles

Short Range: Resolve a short-range missile attack with the total number of missiles

Short Range and Torpedo attack: Resolve a short-range missile attack and then resolve an attack as per 17.11 'Submarine attacks on Task Forces' immediately thereafter if desired by the attacker.

Coordinated Missile Attacks

Submarines in the same zone attacking the same target may coordinate their attack [16.2.2]. For each additional submarine attacking, apply the coordination modifier. If an attack is made, resolve all the missiles together as a single attack.

17.14 Surface Ship Launched Missiles

Soviet Surface Units with black or yellow missiles represent long-ranged anti-ship missiles such as the 'Shipwreck' (yellow) and 'Sandbox' (black). They may be fired at a 'Good' detected Task Force in the same zone as a Soviet Task Force. No roll is needed.

A 'Poor' detected target may be attacked if the Soviet TF is fast on a roll of 6+.

If the roll is failed, the OPS for the action are still expended.

If the target Task Force of a Soviet yellow or black missile attack by surface ships has any F-14 fighters within it, roll one die if the Soviet TF is 'Poor' detected or two if 'Good' detected. Any rolls of 1 or 2 kill that many missiles prior to SAM resolution, selected by the NATO player.

R#1

White, blue and red missiles may only be used through the play of cards except for in the Baltic Sea. (where ranges are likely to be much shorter than elsewhere). In the Baltic Sea, a Surface Unit may fire white and/or red missiles at a 'Good' detected Task Force in the same zone. Only a single unit may fire from each attacking Task Force.

Separate Task Forces may not coordinate their fire at all with any type of attack. Nor is any type of coordination possible between air, ship or submarines.

Design note: I consider it too early for any real coordinated activity to take place. In this period data-linked command and control systems and computing power required were not yet up to the task.

17.15 Air Launched Missiles

Once all air transit and Interception / CAP attacks have been resolved, remaining missiles are launched at the Task Force. No roll is needed, and all firing units coordinate automatically.

Note: Both sides possess air-launched missiles. The Soviets have very long-ranged cruise missiles and NATO has short-ranged sea skimming missiles such as 'Harpoon' and 'Exocet'.

17.16 Resolving Missile Attacks

Once the number of incoming missiles is known, place the 'Missiles' marker on the Missiles Track, using the correct color(s) to track the incoming missiles. A Task Force may fire more SAMs than there are incoming missiles but this is limited based on the number of incoming missiles (due to ammo conservation and the math involved in large scale anti-missile defense).

SAM Resolution

1) Count the # of SAM points your defending force has and use the following table to determine how many can be used if there are more SAMs available than incoming missiles.

Missiles Fired	SAMs Usable
1-5	# missiles +1
6-10	# missiles +2
11-15	# missiles +3
15-20	# missiles +4
21-25	# missiles +5
And so on, as needed	

2) Roll one die per SAM fired, applying the modifiers and totaling the hits scored. Any missiles not shot down make it through the missile defenses – proceed to Missile Resolution [17.17].

Where there are different types of missiles, the TF owner can choose SAM targets as desired.

SAM Hit DRMs	
Shooting at NATO missile	-1
Shooting at Soviet red or yellow missile	-2
Shooting at Soviet white submarine launched missile (slow, high) (SS-N-3 Shaddock)	+2
Bad Weather	-1
Submarine short range missile launch and not yellow missiles	-2
Soviet SAM Technology marker	+1

SAM Hit Roll	
1-2	Miss
3-8	1 Hit
9-10	2 Hits

17.17 Anti-Ship Missile Resolution

If any anti-ship missiles are left after SAM resolution, resolve this step to decide whether any ships are hit. Roll one die per 2 remaining missiles, and one die for a remaining single missile, which cannot hit more than once and misses on a 1-2.

Incoming Missile Resolution			
Rolling one die per 2 missiles remaining, and a single die for an odd remaining missile. Apply DRM's			
2 Missiles		1 Missile	
DR 1-7	1 Hit	DR 1-2	Miss
DR 8-10	2 Hits	DR 3-10	1 Hit
DRMs	Good Detection	+1	
	Bad Weather	-1	

Once the number of hits is known allocate the hits according to the following:

Roll two dice, and if either die shows a 9 or 10, the first hit is allocated by the attacker. Otherwise:

NATO attacking allocates every 2nd hit (2,4,6,8,10...)

Soviets attacking allocate every 3rd hit (3,6,9,12...)

Soviets may resolve as NATO (allocating every 2nd hit) if they have the 'Missile Technology' marker.

Effect of Hits

Each hit represents about six weapons finding targets, somewhat more than a torpedo hit.

Non-Capital ship units which are hit take a step loss.

Amphibious units and Convoys take two hits per hit scored.

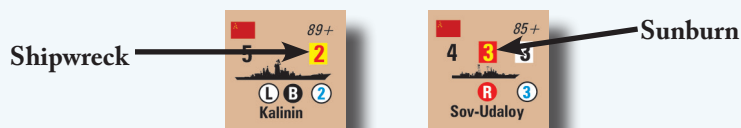
Capital ship units follow the process in [14.1] per hit scored.

Determine the effect of all hits on capital ships after allocating all hits.

17.18 Shipwreck (SS-N-19) and Sunburn (SS-N-22)

Several Soviet Surface Units have 'Shipwreck' (yellow) missiles and 'Sunburn' (red) missiles. NATO must apply the modifier for shooting them down with SAMs.

When different missile types attack the same Task Force, you'll need to keep track of each type individually. They all attack together at the same time and the NATO player can choose which missile type to target with SAMs for each roll of the die, before the die is rolled. Some additional Missiles counters are provided for the Missile Track to help with this book-keeping.



Design note: These weapons were the first Soviet sea skimming missiles. Due to this, and other new capabilities (built in jammers and 'swarm' attack features) these missiles benefit from a large modifier when engaged by NATO SAMs. It's just a shame there are so few of them!

17.19 Attacks Against Land Targets

Land targets can be attacked by cruise missiles and bombs. Both sides have dedicated land attack cruise missiles which have a green background and can only be used against land targets.

Most Soviet missiles can be used as cruise missiles against land targets. Only Soviet white and red ship-launched missiles are not usable in this role. Otherwise, Soviet missiles can be used at range 0 (within the same zone) against land targets. Note that Soviet white submarine launched missiles can be used against facilities, as they represent a longer-ranged weapon.

NATO strike aircraft not equipped with green background cruise missiles must use bombs to attack land targets. They cannot use blue missiles as these are anti-ship only.

Design note: 'Bombs' also represents 'Walleyes' and 'Mavericks', but the range is such that they don't get any bonus beyond the effectiveness bombs already grant.

Any Strike unit with missiles must use them when attacking unless attacking troops' [17.24].

To summarize:

All units with green-background missiles have land attack cruise missiles.

All NATO units without green-background missile must use bombs

All Soviet units may use their missiles as cruise missiles except ships with white and red missiles

17.20 Aircraft Bombing Missions

Once all air transit and CAP attacks have been resolved, air Strike units can resolve their attack. Each step of an air Strike unit receives one bombing die.

SEAD (SAM Suppression)

Striking air may allocate one or more available bombing dice to suppress or destroy enemy SAM sites. Two US CAGs together also receive a bonus SEAD die whether at one or two step strength. *Design note: It is assumed the Prowlers stay out of the way in relative safety and perform defensive jamming and anti-radiation attacks.*

Allocate the dice to roll for SEAD and then roll all dice together.

Mark permanent damage with Hit markers.

Each hit reduces both values on a SAM counter by one.

SAM Suppression (SEAD)	
Die Roll	Result
1-2	Miss
3-5	-1 SAM value for this attack only
6-8	-1 to SAM value (place a hit)
9-10	-2 to SAM value (place 2 hits)

SAM Attacks

Once SAM suppression has been resolved the player owning the target facility may roll with remaining SAM dice indicated on the counter. (the leftmost number vs. strike aircraft)

SAMs vs bombing

Each 6+ removes one bombing die and a roll of equal or higher than an air Strike units' Defense value causes a step loss as well as removing the die of the just shot down step. Apply +1 DRM if the Soviet has the SAM Technology marker.

Bombing: Roll the remaining bombing dice and add modifiers. If splitting between targets in the same zone, split before rolling any.

Example: A US CAG attacking the Kola Peninsula may allocate one die to attack the airfields and another die to attack the ports.

Roll	Bombing Result
1-3	Miss
4-8	1 hit + 1 collateral
9-10	2 hits + 2 collateral

Bombing modifiers	
Bad Weather	- 1
Some range unspent (carrying more bombs)	+ 1

17.21 Cruise Missile Attacks

NATO Cruise Missiles (Tomahawk 'TLAM') Green-Background Missiles

Unlike other missiles that must be launched from a Sea Zone adjacent to the target Land Area, TLAMs can be fired ONE Sea Zone away from the target's Sea Zone.

Example: *The Soviet Baltic Sea facilities could be attacked by TLAM from the North Sea, North 3-4.*

The green background number is the number of cruise missile dice available.

Important! Tomahawk is first available in 1985 and are not usable at all if the scenario takes place in 1983, despite the counter showing their presence.

Design note: *At this point they are all nuclear warheads and/or anti-shipping variants which are already represented via card events.*

Soviet Yankee Notch and card-based Tu-95 (AS-15) Green-Background Missiles

Unlike other missiles that must be launched from a Sea Zone adjacent to the target Land Area, AS-15 can be fired TWO Sea Zones away from the target's Sea Zone.

Example: *Norfolk NS in the USA could be attacked from Mid-4 or Mid 8-9.*

The green background number is the number of cruise missile dice available.

Soviet Missiles

All Soviet ship-launched black and yellow missiles, all non-green submarine missiles and all Air Unit missiles can also be used to attack land targets. These must be carried to the target zone. Count the total missile points being fired and convert on the following table:

Soviet Missiles vs. Land Targets	
Missile Points	Dice
1-3	2- low
4-7	1
8-12	2
13 or more	3

No coordination is possible between submarines, air and ship units – each type of attacker computes and rolls separately. Multiple units of the same type in the same zone automatically attack together.

Note: "2-low" – Means roll two dice and use the lower roll.

Cruise Missile SEAD (SAM Suppression)

Cruise missile dice may be allocated to suppress or destroy enemy SAM sites. Allocate the dice to roll first and then roll all dice together using the process at [17.20].

SAMs vs Cruise Missiles

Roll the number of dice indicated on the SAM counter (bottom number) modified for damage taken.

An 8+ shoots down one point of enemy cruise missiles.

A 6-7 degrades a point of cruise missiles, applying -2 to its roll when attacking.

Apply +1 DRM if the Soviet has the SAM Technology marker.

Cruise Missile Strikes

Roll the remaining strike dice. If splitting between targets in the same zone, split before rolling any. Apply all cruise missile attack modifiers.

Roll	Cruise Missile Result
1-3	Miss
4-6	1 hit
7-9	1 hit + 1 collateral
10	2 hits + 2 collateral

Cruise Missile Attack Modifiers	
Bad Weather	- 1
Cruise missiles attacking Troops markers	- 1

17.22 Collateral Damage

Collateral Damage

Collateral damage causes step losses to units at or flying from the airbase or port under attack. The bombing or Cruise Missile Results Chart give the # of collateral step losses which must be applied.

Total all collateral step losses caused before applying any

Even losses are allocated by the defender. Odd losses by the attacker.

Example: *Keflavik on Iceland is hit by missiles and takes 2 collateral damage. Two step losses will be allocated among units flying from Keflavik, even if they are currently flying an Interception in a different zone and even if they just fought in air to air combat (some of the squadron is on the ground).*

17.22.1 Soviet Air Complexes

At Kola, Baltic and Black Sea air bases NATO may force air collateral damage to be applied to Bear, Badger and Backfire aircraft (of any types) or to choose to target fighters (they are separate airfields). NATO may also choose to target both types of airfield by allocating collateral hits to both targets.

Additionally, **one** collateral hit may be allocated (by either player) to remove a RORSAT temporarily from the game, returning three turns from the current turn (RORSAT is selected by the US player on a die roll of 6+, otherwise by the Soviet player).

Design note: *This represents targeting of communications facilities.*

17.23 The Kola Peninsula

For each hit scored against the Kola Peninsula facilities, the Soviet player receives one First Strike Point.

Design note: As one of the most important military bases to the Soviets, an attack against Kola would be likely to escalate the situation quickly and perhaps uncontrollably.

17.24 Striking Soviet Troop Markers

NATO may attack Soviet troop formations to slow the Soviet advance. The main target are the troops printed on the map near the Yugoslavian border which affects the Europe South War Track, but Soviet troops in Norway and Denmark can also be attacked to affect the relevant Invasion Track.

When flying to Norway, if the current Invasion Track position is the Bodo box or above (between the Bodo box and the start box) then Air Units must fly to the Barents Sea North 9-10 to strike the adjacent Land Area (Northern Norway) otherwise to North 5-6 to strike troops in Southern Norway.

(The Invasion Track has a N and S designation as a reminder).

If there is a Soviet airbase in the attacked location, fighters there fly CAP against the strike as well as the defending fighter rule below.

When striking, the NATO player may take escorting fighters, with the following system representing Soviet fighters and SAMs (there are no counters). The NATO player rolls attack dice with escorting fighters, adding the units Tactical Value and aiming for a target number of 8 or higher. Cross reference the result on the table below, which shows how many attacks the Soviet player can roll against each Strike unit.

For each Soviet attack, roll a single die. On a 6+ one step of bombers does not attack, and on a roll equal to or higher than a Strike unit's Defense value, also lose a step of strike aircraft.

On a natural 10, the Soviet player may instead choose to cause a step loss to a Fighter unit.

Apply +1 DRM if the Soviet has the 'Air to Air Missile Technology' marker.

Striking Soviet Troops	
Hits	Soviet Attacks
0-1	2 dice vs each striking unit
2-3	1 dice vs each striking unit
4+	1 single die

Cruise missiles can be used against these targets (represents attacking HQ's, depots, bridges etc.) – apply the modifier for attacking troops within the cruise missile attack process.

Example: A US Task Force launches a strike against the troops in Yugoslavia with 2 x CAG and 1 x F-14. The F-14 rolls 4 dice, adding +2 Tactical Value to each. 3 dice score 8 or higher. This results in one die versus each striking unit. The Soviet player rolls 1 die against each CAG. A 6 causes one CAG step to not be able to attack and 9 shoots down one CAG step as well as preventing their attack. – two steps remain to complete their bombing mission.

18.0 AMPHIBIOUS LANDINGS & NATO TROOP DELIVERY

Amphibious Landings can be resolved by any force which is marked 'landing' when the force is eligible to move again and the force occupies an eligible Amphibious Landing zone on the map.

Note: NATO only ever makes Amphibious Landings against a Soviet occupied Iceland – otherwise NATO always delivers troops to ports instead. As such port damage and mining may affect them but there is no Amphibious Landing resolution.

Any submarines in port at the location being invaded may move to sea immediately and may roll to attack the force using [17.11] Submarine vs Task Forces attacks. For each submarine rolling to attack, expend one Operations Point of the owning player. Mark each "Spent".

Following this, complete the Amphibious Landing steps:

Determine the number of dice to roll by cross referencing the landing location and current turn.

Location	Turn 1-2	Turn 3-4	Turn 5 +
Norway	4 dice	3 dice	2 dice
Roll one less die if landing at Bodo/Narvik			
Denmark	6 dice	4 dice	3 dice
-2 dice if Soviet troops ashore in landing location			
-1 die if the landing location is lightly or heavily damaged			

Roll all the dice and then resolve starting at all '1's, then all '2's and so on.

Each die causes an effect which can be cancelled if the Soviet player has the relevant capabilities within the Task Force. No capability can be counted more than once.

Die Roll	Effect
1-3	Fast Attack Craft Cancel by expending three white or red missiles (red only vs Denmark). For each not cancelled, change the die to a 6 and resolve later.
4-5	Ships Cancel by expending two black or yellow missiles. For each not cancelled, change the die to a 6 and resolve later.
6-8	Missile/Air strike Cancel by counting a total of four SAMs which cannot be counted again. Cancel one for free if TF is not marked with a Good Detection marker.
9	Mines Cancel by taking a step loss on any ship
10	Land Forts and Artillery Cancel one if Sverdlov is present.

Hits	Troops
1-2	2
3-4	1
5+	0

Any remaining dice cause one Hit marker on any Amphib, selected by the NATO player.

Once completed, determine the number of troops to land by cross referencing the current hits on each Amphib unit and place the troops onto the Invasion Track at the landing location. Resolve the capture of airfields and then remove the Amphib unit(s) from the map.

18.1 NATO Troop Delivery

NATO delivers troops to ports and so does not use this process. Instead, apply port damage and mines [13.2] and then 18.3, below

18.2 Soviet Overland Troops to Norway

The Soviet player receives troops as reinforcements which go to Norway – they are placed in the currently occupied box on the Norwegian Invasion Track when they arrive.

18.3 Troop Placement

If NATO delivers troops at Oslo, land only one troop unless Soviets are in the Orland box.

Design note: These troops are far from the front and will take some time to get into contact with the enemy.

Otherwise, reference the Troops Landing Table and place the correct number of troops.

If there are enemy troops present, first remove them on a one for one basis and then place remaining troops on the track. Note that Iceland has a special rule (21.4).

Example: *There is one NATO troop on the Norwegian Invasion Track. The Soviets invade Bodo and manage to land with 2 hits. This results in the placement of two troops. The first removes itself and the NATO troop. The second one is placed on the Invasion Track at Bodo and then the capturing facilities process is followed to determine Bodo's status.*

19.0 CAPTURING FACILITIES

There are several facilities which can be captured during the game: Iceland, Bodo, Orland, Kiel/Schleswig, Antwerp/Amsterdam, La Spezia, Toulon/Istres.

The majority of these are captured automatically if a Soviet Army or Troops marker moves into the relevant box on the War or Invasion Track. Place a 'Heavy' damage marker on the named facility if this happens and rebase any NATO units there to the nearest base measured in Sea Zones; selecting if there are multiple options.

Keflavik, Bodo and Orland can be captured due to an Amphibious Landing - If an enemy Troop marker is placed at any of these sites, place a Soviet Airbase marker and roll for the condition of the airfield: roll one die per troop landed and reduce one level from 'Heavy' to 'Light' to 'undamaged' for each roll of an 8 or higher. Place the appropriate number of hits on the facility so that it is damaged to the correct level.

Then rebase any NATO aircraft there to the closest friendly airfield, all are spent.

The Soviet player may immediately place up to two Fighter units and two Strike units at the new base from the Kola Peninsula bases, Baltic bases or another captured airfield. Any Strike units moved are "Spent". A facility always retains at least the damage it had when captured. If the captured site is Orland/Trondheim, Bodo/Narvik is automatically captured as well but it is always marked 'Heavily' damaged with no roll allowed to reduce this.

Note: The capture of La Spezia gives the Soviet player a port with a damage value of 2/4. It has no effect on the operation of Toulon/Istres.

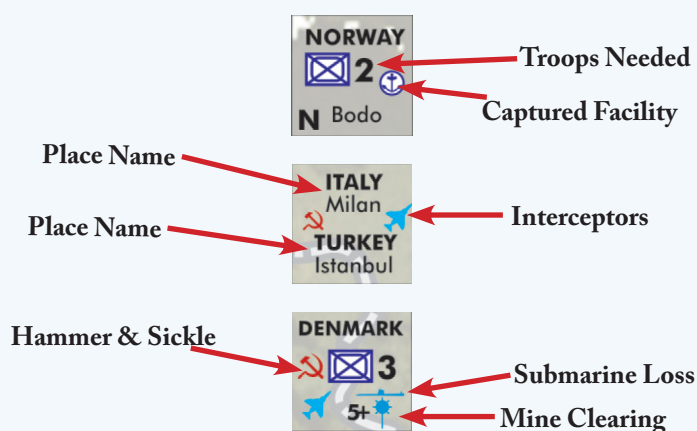


Abandoned Soviet Submarine base in Russia

20.0 WAR & INVASION TRACKS

The game map has several tracks which show the status of the Soviet offensive into Western Europe. Overlaying Germany are the Europe North and South War Tracks and the Danish Invasion Track with the Norwegian Invasion Track for Norway.

At the end of each turn, check each track to see whether the fronts advance.



Invasion Tracks (Norway and Denmark)

Compare the number of Soviet Troops on the track to the 'Troops Needed' value. If the number of Troops is equal or more, advance the track one box. If the number of Troops is double or more, advance one box and then make the check again in the new box to see if the marker advances another box.

NATO may strike Soviet troops to stall an advance for the current turn.

One advance is cancelled for every three hits present.

Remove up to three Hit markers at the end of every turn.

Occupation of Kiel/Schleswig

If the Soviet player occupies the Europe North War Track in the 'Hamburg, Munich' box or further west, they may forgo one box of Western movement and instead occupy 'Kiel/Schleswig'. This has the effect of capturing the facility [19.0] and places two additional Troop markers onto the Denmark Invasion Track. Also move the Invasion marker into the space the arrow points to – the fourth box along, unless it has already passed this space.

War Tracks

The Europe North and South War Tracks move one box West each turn.

An advance is cancelled if the NATO player removes one 'Supplies' or two 'Partial Supplies' markers from the War Track. This is never mandatory – the supplies remain until NATO chooses to use them.

If there are three Hit markers on the Europe South War Track, this cancels an advance.

Then remove up to three Hit markers at the end of every turn.



Track Icons

Interceptors: Are shown on the Danish Invasion Track and the Europe South War Track. Each shows the number of interceptor's active over Denmark and Turkey respectively for purposes of Interception [17.2].

Submarine Loss: Represents the loss of ports and related facilities. When a Front marker occupies a box with one or more symbols, NATO must remove Submarine steps if it is possible to do so at the end of the turn. Norwegian losses must be Norwegian. German must be German. Danish may be Danish or German. All selections are made by the NATO player.

Mine Clearing: Allows the Soviet player to attempt to reduce the current Mining level of the Danish Strait at the end of the turn by rolling the indicated value on one die [21.2.1]

Captured Facilities: These are always represented by a port icon. Whenever a Soviet army occupies one of these boxes, a NATO facility has been captured [19.0].

Hammer and Sickles: This shows when the game is won. To win the Soviet player must be able to count four hammer and sickle symbols on War Tracks overrun by Soviet armies.

Nukes - Entering the Ruhr: NATO will use nuclear weapons if the Ruhr box is entered and the Soviet player holds less than two First Strike Points or less than four First Strike Points if chemical weapons have been used. If nukes are used, place the War marker one box East of the Ruhr, then remove all supplies from the War Track. NATO's first use of nuclear weapons prevents the Soviet player from triggering the use of battlefield nuclear weapons for the rest of the game.

If the Soviets have already used battlefield nuclear weapons then this NATO effect is ignored (only the first use player gains an advantage) however, if the Soviets enter any Hammer and Sickle space via the use of battlefield nuclear weapons, reduce its worth to one Hammer and Sickle – this can be marked with the Nuclear Weapons markers as a reminder.

21.0 NATIONS

21.1 Turkey



Turkey has a Mediterranean Convoy destination port to which NATO may deliver supplies. It is also a barrier to Soviet aircraft—they may overfly Turkey but will be intercepted by the number of aircraft shown by symbols on the Europe South War Track. Use the Turkish F-4 Phantom printed on the map for all Interception resolution.

The Turkish Strait is closed for the duration of the game.

Design note: *It is assumed it's blocked off either by the Turks themselves or Soviet bombing.*

21.2 The Danish Strait

The Danish Strait may be 'Lightly' mined or 'Heavily' mined depending on the scenario.

Increase the Mining level by one at the end of each turn if the Soviet player has not placed a Troops marker on the Danish Invasion Track via Amphibious Landing at some point during the game.

Any Soviet submarine moving through must roll one die if 'Lightly' mined or two dice if 'Heavily' mined, taking a step loss on rolls of 1 - 4. Roll one die less if the Submarine Unit is marked Small (S) or Non-Magnetic (N) to a minimum of 0 dice.

Soviet Task Forces roll two dice ('Light') or four dice ('Heavy'). Odd hits must go to a Convoy or Amphib unit if present. Other hits are allocated freely by the player taking the damage.

21.2.1 Clearing the Strait

At the end of any turn when the Soviet troops marker on the Danish Invasion Track is in a box with a Mine symbol the Soviet player may roll the die to reduce the strait mining value by one level (Heavy to Light to Clear). The required roll is listed on the track.

If the Strait is cleared then French, British, Belgian and Dutch ports are immediately marked 'Heavily' mined. **Design note:** *Soviet fast attack craft surge into the English Channel.*

21.3 Norway



If the last box on the Norway Invasion Track is entered, Oslo is occupied and Norway surrenders.

Thereafter, at the end of each turn, including the turn of surrender, the Soviet player may choose one of the two following bonuses:

1. Roll a single bombing die against the UK airfields, or against the Belgian/Dutch facilities
2. Place a 'free' Troops marker into the Danish Invasion Track on a die roll of 7+

21.4 Iceland



A Soviet Amphibious Landing on Iceland will capture Iceland unless garrisoned by NATO troops which have been delivered by a NATO Amphibious unit or by a pre-game optional event.

Capturing Iceland disables SOSUS permanently in the East 9-10 and East-6 Sea Zones and provides a Soviet captured airfield.

Either side landing on Iceland delivers troops normally as per an Amphibious invasion, but there are no invasion defenses. Delivered troops remove one another on a one for one basis.

If Sverdlov or N Jersey/Iowa is present, and the last friendly troop would be removed, retain one troop on a die roll of 4+ (Iowa) or 6+ (Sverdlov).

A facility is captured only if a friendly Troop marker remains after the combat. If both sides are mutually destroyed by a landing, place 'Heavy' damage on Keflavik.

21.5 Cuba



Cuba begins as an inactive Soviet ally. This allows limited Soviet basing on Cuba, and via card play, participation of the Soviet Lourdes intelligence station and Cuban intelligence apparatus which the NATO player cannot influence unless NATO chooses to attack Cuba. Additionally, Cuba has a minor port which can rearm Soviet submarines. No Soviet air Strike units may base in Cuba while inactive, but one MP unit may base there. The Lourdes facility begins each game on Cuba. (See 13.14 for the effect of destroying the facility).

Design note: *The largest foreign Soviet intelligence-gathering facility in the world during the Cold War, located close to the US coast – the station was extremely important to Soviet intelligence gathering operations.*

The US may choose to attack Cuba during the game by spending six OPS and resolving the following:

1. The Soviet player deploys the Cuban submarine within one zone of Cuba on the roll of 8+ otherwise it begins in port. Deploy the Cuban Air Units on Cuba with the SAM marker.
2. The US may resolve up to three OPS worth of Submarines or Task Force cruise missile attacks against Cuba for no OPS cost.
3. Place the four US units marked 'Cuba' in Florida and fly an immediate Strike mission with them.
4. Cuba is now active, the Soviet player may use Cuban forces as his own, as well as immediately deploying up to three air units available at any Soviet base to Cuba. All are marked 'Spent'.

21.6 Syria



Syria starts as a Soviet ally but has no forces of its own. It is a Soviet base.

21.7 Libya



Libya is neutral and only enters play via an option [29.0].

22.0 SOVIET BOOMERS (SSBNs)

Design note: A 'Boomer' is a slang term for an SSBN submarine which carries nuclear missiles and is designed to provide an assured nuclear strike capability against the enemy.

Each SSBN in the game has a Payload value on the counter which represents the throw weight of that unit in terms of warheads, TNT equivalent, and targeting capability.



SSBN Payload Value

The Soviet Boomers are regular game units which can operate anywhere on the map but have three specific areas where they are 'on-station' and can provide a game effect: The Boomer Bastion, under the Arctic Ice and on-station at First Strike Zones [6.4 – 6.6].

22.1 First Strike Points (FSP's)

Design note: First Strike Points represent the nuclear posturing of the two sides. If the Soviets can maintain a credible First Strike capability, the Politburo (Soviet leadership) will feel able to take aggressive actions such as using nuclear weapons or assassinating high value targets. In game terms, FSP's can be spent on powerful effects for the Soviet player.

The Soviet player earns First Strike Points through card play. In addition, at the end of a turn marked with a First Strike icon, if the relevant SSBN submarine type is in an appropriate First Strike area then count the total Payload value on-station. If the total is 6 or more, give the Soviet player a First Strike Point.

In 1989 a Yankee class is still worth two per step for the Soviet player, they are just worth less to NATO when killed. This is signified by the '-1 per step' marker placed during setup as a reminder.

Design note: By 1989 the Soviets knew the Yankee was heavily outclassed, but the Soviets rarely threw anything away that was still sea worthy, so patrols continued.

First Strike Points can be 'Spent' when active on the following:

- Anti-Satellite System - place one or two points in Tyuratam. While present, prevents NATO play of 'Whitecloud' and 'Snoopers Detect Raid'. Causes each NATO detection die roll against Soviet Task Forces to apply -1 per First Strike Point in the Tyuratam box.

Additionally, if there are two First Strike Points in the box, NATO must deduct 1 OPS from each card played. At the end of each turn roll one die and remove a First Strike Point on a roll of 6+.

- Take a step from the strategic reserve – one point per step.

If taking one step, reduce a unit in the reserve by a step and increase a unit on the map by a step.

- Trigger the use of nuclear weapons at sea - three points [26.0].

- Use battlefield nuclear weapons - five points (move an army two boxes, reducing victory locations to one Hammer and Sickie. Move Soviet stability one right. Prevents NATO use of nuclear weapons [20.0]).

- Prevent NATO use of battlefield nuclear weapons when entering the Ruhr space - two points or four if chemicals were previously used by the Soviet player.

22.2 Soviet Nuclear Strike Capability

Design note: If the US can eliminate Soviet second-strike assets the Soviets will begin to fear a nuclear exchange.

Keep track of the SSBN Payload value killed on the Soviet Nuclear Strike Capability Track –use the SSBN Kills marker on the Missile Track or dead SSBNs and Hit markers to keep a tally.

Note: Killing one step of SSBN earns the difference in value between its two-step and one-step side.

Example: A two-step Typhoon has a Payload value of 8. If NATO kills one step, add 4 to the Payload value killed.

When the Payload value killed falls within the requirements listed on the Soviet Nuclear Strike Capability Track, move a box up on the track and roll the indicated number of Soviet stability dice.

These rolls can affect the Soviet Stability Track on the map which has a variety of negative consequences for the Soviet player including an immediate game-ending coup.

23.0 NATO BOOMERS (SSBNs)

NATO Boomers are assumed to be dispersed over the world's oceans and are not represented by specific counters.

Design note: Unlike NATO the Soviets really didn't have the capability to hunt NATO SSBNs in any significant way. Additionally, the SSBNs were generally the best technology available and spent their time trying to hide and moving slowly. They would have been very difficult to detect and attack in any number.

The 'SSBN Hunting' marker represents the Soviet effort to hunt down NATO Boomers. It can be earned by the Soviet player through spy activity or through the roll of a '10' with an Anti-Submarine Warfare die [17.8]. Once in-hand it can be spent through the roll of another '10' when resolving Anti-Submarine Warfare. This then results in a roll on the NATO Boomer Hunt Table.

Die Roll	NATO Boomer Hunt
1-3	Counter-Attacked The Soviet submarine takes a step loss
4-7	Near Miss The Soviet player gains one First Strike Point.
8-10	Sunk! The Soviet player gains two First Strike Points or one First Strike Point and reduces the Soviet SSBN killed total by three points.

Add the Tactical Value of the resolving submarine to the die roll.

24.0 SOVIET SPIES

The Soviet player has a 'Spies' Track' in the top left of the map which represents the thoroughness of Soviet infiltration of NATO agencies.

The Soviet player may use spies for the following actions:

- Place a Poor Detection marker on any NATO Task Force.
- Take the SSBN Hunting marker.
- Re-roll up to three ASW dice rolled. One re-roll per unit maximum.
- Automatically succeed on all First Strike submarine evasion rolls in a single zone (*even after rolling*) [6.7].

They can be used whenever desired until there are none remaining. After each use, roll a die and reference the Soviet Spy Infiltration Track for the die roll range marked in the current box. If the die is within the indicated range, a spy has been caught - move the track down one box (to the left).

25.0 PRE-GAME TECHNICAL THEFT

Design note: In the Cold War the Soviet intelligence apparatus specialized in stealing NATO technical secrets and equipment. Soviet intelligence efforts were extensive and extremely successful, with the GRU (Army Intelligence) maintaining a shopping list of NATO secrets to steal next and having teams dedicated to copying and reverse-engineering all manner of stolen technical items.

In the game NATO generally has the edge as its units are better rated.



The Soviet player can choose to put in play some of the available Technology markers at the start of the game, each of which has the following effects: Note that only Soviet units gain any bonus.

SAMs	Cost 2 spies
Add +1 to all SAM to-hit rolls (land and sea based).	
Anti-Ship Missiles	Cost 1 spy
Resolve Soviet missile hit selection as though NATO – allocating every 2nd hit instead of every 3rd hit.	
ECM	Cost 2 spies
Soviet Bear-G and Tu-22M Backfires always receive an 8+ save to avoid a step loss when airborne. Ignored if they already receive a better save.	
Torpedoes	Cost 2 spies
Apply +1 Tactical Value (in total) whenever a Soviet nuclear submarine attacks a NATO submarine. Soviets may use the Harpoon rule when attacking NATO Task Forces. <i>Soviets develop a similar capability weapon – AS-20 Kayak.</i> [17.11, step 4]	
Air to Air Missiles	Cost 1 spy
Apply +1 when attacking with fighters, and when resolving 'Striking Soviet Troops' [17.24]	

Stealing Technology

The cost of each Technology marker, in spies removed from the Spies' Track, is given on each marker.

26.0 NUCLEAR WEAPONS AT SEA

These are triggered via the use of First Strike Points [22.1] When in use apply the following rules:

Note: Using battlefield nuclear weapons does not trigger the use of nuclear weapons at sea. Also note that NATO receives no benefit for the use of nuclear weapons at sea, but there are cards which can punish the Soviet player for first use (see 'Nuclear Fear' NATO card)

Soviet Shkval: (nuclear counter-fire super-cavitating torpedo)

A Soviet nuclear submarine taking a step loss caused by a NATO submarine's N10 roll also causes a step loss to the attacking submarine.

Design note: A very fast nuclear torpedo designed as a revenge weapon for Soviet submarines

Soviet Nuclear Torpedoes

All Soviet submarines automatically use nuclear torpedoes when attacking a Task Force.

Any hit Capital ship with a damage modifier value is automatically destroyed. Capital ships without a modifier follow the rules in [14.1].

Non-Capital ship units follow the standard rules (*they are dispersed*).

Convoys and Amphibs take two hits on the first 7-9 result (hit on PM/Amph/Convoy) instead of the usual one hit. This applies per attack.

Soviet Nuclear Anti-Ship Missiles:

All Soviet missiles automatically use the following rule when attacking a Task Force:

For every three incoming missiles, the third one is nuclear armed, so if three missiles make it through the SAMs one is a nuke but if only two make it through, neither is a nuke. Resolve nukes first, then resolve non-nuclear missiles.

Note: Nuclear weapons always hit something.

Roll a die, on an 8+ the Soviet player allocates it, otherwise the NATO player allocates it.

Any hit kills the target ship entirely, except for a Convoy or Amphib unit which takes three hits.

Design note: This represents ~4 high yield tactical nuclear weapons.

Soviet Nuclear SAMs

If the Soviet SAM level of a target land facility is at least two on the counter, treat it as being worth +2 more than its current printed value against Air Units only (not against cruise missiles).

27.0 SOVIET STABILITY

The Soviet Stability Track shows the stability of the Soviet bloc during the conflict. It can be altered via the play of certain card events and through the sinking of Soviet SSBN Payload value. When the Soviet Stability marker moves to the right, immediately apply the effect detailed.

Politburo Intervenes: Causes the Soviet player to take the 'Nuclear Strike' card from the deck (reshuffle after) and place it in-hand. From this point on it cannot be discarded nor played for its OPS value. It can only be played for the event and it must be the card held at the end of the turn.

28.0 NATO LOSSES

The NATO losses marker starts the game off-board. Use the Turn Track as a convenient place to record NATO loss points earned. During the game if the NATO losses marker ever reaches 6 points, the Soviet player scores 1 Hammer and Sickle (of the 4 needed to win the game).

Every third Convoy Massacre earns the Soviet player one NATO loss point.

1 point is earned for damaging a US carrier, or 2 points if the US carrier sinks. If a US carrier is north of the SOSUS line at the time of its damage or sinking, this is worth one additional point.

Damaging a carrier twice or more has no further effect – it must be sunk to earn the additional point.

Design note: This represents US fear of losing the carriers and the international response to such a thing happening. The bonus for operating north of the SOSUS line is because in so doing the US clearly exposes its main offensive arm and is poking the bear. A congressional hearing awaits the commander who loses his carriers in the Norwegian Sea.

29.0 OPTION MARKERS

An option is an alt-history event or technology that could have happened but historically was never deployed.

If using this rule, follow this process before starting the game (*You'll need an opaque container*):

- The US player draws two Option markers, selects one and returns the other to the cup.

For each additional First Strike marker given to the Soviet player (up to two) the US player may draw two additional markers, keeping one more each time. All are revealed to the Soviet player.

- The Soviet player draws two Option markers, selects one and returns the other to the cup. For each First Strike marker returned, or spy expended (two max) the Soviet player may draw 2 and keep 1 additional marker.

SOVIET STABILITY TRACK

Stable	Allies Waiver	Allies Withdraw	Politburo Intervenes	Backfires Withheld	Command Paralysis	Internal Coup
Workers of the World unite!	Roll one die for each War Track and lose this turn's advance on a 1 or 2	NATO places a free partial convoy on any front	Place the 'Nuclear Strike' card in your hand. It must be held or played for the event.	Backfires and Bear-G may only be used against a 'GOOD' detected TF	All Soviet cards are treated as worth two less OPS	The game ends. The Soviet Player counts one less

RESHUFFLE THE DECK

SOVIET STABILITY DIE ROLL

Die Roll	Result
1-5	Lose 1 FSP OR move one box on the Soviet Stability Track (towards coup)
6-8	Move one box on the Soviet Stability Track (towards coup)
9-10	Lose 1 FSP AND move one box on the Soviet Stability Track (towards coup)

Option Marker	Effect
Cable Tapping	Once per turn count any Soviet airstrike as 'detected by interceptors or re-roll all dice rolled by one unit in Anti-Submarine Warfare with an <i>additional</i> +1 applied.
ASAT System	Twice per game select up to 3 RORSATs, roll a die for each, and remove each on a 6+ then give 1 FSP to the Soviet (each time).
Hydrophones	Add two Hydrophone markers to the map, each provides +1 Anti-Submarine Warfare dice <i>within</i> its Sea Zone. They must be placed in different zones.
Production	Select 8 points worth of optional units to add to the game - Patriot's cost buys both counters, deployed at any NATO base. - UK Queen Elizabeth replaces Invincible/ Illustrious and Air Units. - Norwegian FAC flotilla adds 2 dice set to '1' during all Soviet Amphib landings in Norway. (Place in Norway) - Canadian Hydrofoils see rules to right .
Helicopters for Convoys	Each NATO Convoy with 3 hits or less counts as having an accompanying helicopter carrier for submarine vs. Task Force attacks.
Decoys for Convoys	Amphibs and Convoys gain an 8+ save against any hits caused by Soviet torpedo attacks. Nuclear torpedoes ignore this rule.
Reinforce Iceland	Iceland may receive any two of the option below: Additional F15 - air unit;  Patriot battery - A chance to shoot down attacking cruise missiles.  Base Hardening- makes it harder to heavily damage the base  One Troop marker - Adds US troops to the defense of Iceland. 

Sea Lance	US submarines are equipped with Sea Lance rocket torpedo (allows US use of the rocket torpedo rule by all US submarines).
Tomahawk	Add 4 points of conventional cruise missiles to the order of battle, two may be submarine launched from 1985. All must be added to US units only. A Submarine Unit can only have one cruise missile in total. A Surface Unit can have a maximum of two cruise missiles in total. In 1983 these are the only conventional cruise missiles NATO will have.
Greece and Yugoslavia active	Add the Greek Submarine Unit to the game beginning in either Med 5-6 or Med 7-8 at the NATO players option. Place the option marker on the Europe South War Track. At the start of Turns 1 and 2, roll three dice and place one hit on the Soviet troops in Yugoslavia for each roll of 7 or higher

Canadian Hydrofoils move when Slow Ships move (moving one zone) and may spend one Operations Point to roll their ASW dice, marking themselves 'Spent'. They become Fresh at the end of each turn.

They can take a step loss only by missile attack requiring one or more units in their zone to expend 3 missiles to cause a loss. They must stay within 1 zone of CFB Halifax at all times.

Thanks to Douglas Burrell for the suggestion. Further reading: 'RCN FHE-400 deep water hydrofoil' It was designed as an ocean-going anti-submarine platform but never entered active service.

Option Marker	Effect
C3 Centre	Once per turn upgrade a Poor detection to a Good detection at any time. See below*
SSBN Focus	Add two 'First Strike – half value' markers to the map up to two zones from either US zone. All SSBNs (Deltas and Yankees) in these zones count half value towards First Strike Points calculations. The Soviet player may deploy an additional 2 SSBNs and two escort nuclear submarines to these zones at the start of the game.
Cuba Active	Deploy all the Cuban forces at the start of the game; they are controlled by the Soviet player. Place the Major Port marker in Cuba. Ignore the usual Cuban deployments and deploy up to 4 submarines and any number of ships there. Up to six Air Units may be based in Cuba including fighters taken from any base. The NATO player receives the Cuba units at the start of the game.
Libya Active	Deploy all the Libyan forces at the start of the game; they are controlled by the Soviet player. Up to three Soviet Air Units may begin the game in Libya.

* C3 Centre: Place the centre on the map in either of the Kola, Baltic or Black Sea Soviet facilities. It can be attacked and destroyed which removes it, and it's rule, from the game.

Arab-Israel	An Arab coalition invades Israel with Soviet cooperation. NATO must strike 'War Targets' in/near Israel with twelve strike hits total and/or deliver supplies with 'Med Convoys' with a 'Full Convoy' 'counting as' six strike hits and a partial as four. The Convoy must deliver to a port in Sea Zone Med 7-8 (Israel). If this is not achieved before the end of Turn 6, Israel loses (+2 Hammer and Sickle). The Soviets may base one additional Air Unit in Syria.
Iceland Maskirovka	The Soviet player may place one Soviet Troops marker on Iceland if there are no NATO troops already there. Place all units from Iceland in Scotland and the Soviet player treats Iceland as a captured airbase. The Soviet player must roll for facility damage as though captured by two Troops (<i>a bonus for surprise</i>).
Towed Array and Quieting	Add the 6 markers provided to any nuclear submarines as desired. (<i>This represents earlier acquisition of advanced propulsion and sensor technology</i>).
Production	Select eight points worth of optional units to add to the game. - Imp SAM's cost buys all three markers replacing existing SAM counters. - Akula and Oscar II cost +2 points each in 1983 and +1 point each in 1985.
White Sea Bastion	Soviet SSBNs are protected in the White Sea by mine fields, corvettes etc. Soviet 'Second Strike' card event is not playable. The Bastion is in the White Sea, which is enterable for no additional movement point when entering the Barents Sea – place units into the Bastion box as normal. Bastion defenses now trigger on 1,2 and 3, and ALL Anti-Submarine Warfare rolls there are at + 1 in addition to the Tactical Value. There are no Hydrophones, it's just a very small sea (<i>comparatively</i>).
Turkish Neutrality	Remove the Turkish units from the game and ignore Turkish Interception – Soviet Air Units can overfly the Black Sea to Med 7-8 without risk. Move the Europe South War Track one box left to Trieste. Ignore the 3rd hammer and sickle space on the Europe South War Track (Ankara) – this is earned by occupation of Lyon instead.

30.0 OPTIONAL RULES

"I always find it interesting to know which optional rules the designer plays with, so I have indicated this with an icon next to the ones I like to use"

30.1 ASW Cooperation

Favors neither.

When a MP aircraft is activated and is present in a zone with a Fresh friendly submarine, the MP unit may allow the submarine to roll with two additional dice. Mark **both** units "Spent".

30.2 Harrier and Yak strikes

Favors neither.

Carriers, Yak-38's and Kuznetsov's Carrier Air Group unit can fly air Strike missions following the normal rules. They are armed only with bombs and have a range of 0 (they may attack targets in their zone).

30.3 Bombs Away!

Favors NATO slightly.

All air Strike units can bomb enemy ships - to do so, use the Air to Ground Bombing Table [17.20].

Roll one SAM attack (use the SAMs vs. bombing procedure) for every two points of SAMs the defending Task Force has in total.

A US CAG may perform this type of attack as well as using its missiles - in which case roll one SAM attack per three SAMs the defending force has, resolved immediately after the missile attack. *This represents the A-7 Corsairs bombing whilst A-6 Intruders attack with Harpoon.*

All hits caused by the Strike units are allocated by the attacker and count as missile hits for damage caused (therefore causing two hits each to Convoys and Amphibs).

30.4 Bear-G may not have existed in quantity

Favors NATO moderately.

There is some uncertainty about when the Bear-G was in service from the sources I used. Therefore, in 1983 treat the Bear-G unit as one step (it starts flipped). Give the Soviets two steps total in 1985 and all four steps in 1989.

This reduces Soviet long-range strike capability by three missiles in 1983 and six missiles in 1985. As partial re-payment award the Soviet player two saved OPs at the start of the game or an additional two card draw whenever desired, or a mixture of one of each.

30.5 F-14 Phoenix Missile Defense

Favors neither.

This option allows the NATO player to use F-14's to defend against Soviet missile attacks with Air Units by shooting down missiles instead of trying to attack the bombers. The launching bombers will not be damaged at all by the CAP F-14's.

When under air attack and NATO F-14's are defending, the NATO player can declare a 'Phoenix Defense' and roll the # dice on the



F-14's counter (one F-14 per attacking air Strike unit maximum). For each roll of 1 through 3 on each die, that many missiles are shot down prior to SAM fire.

30.6 Dummy Task Forces

Favors NATO slightly.

A Task Force may be completely empty. If such a Task Force is ever 'Good' detected it is removed from the map – the Task Force is returned to the Task Force display. A 'Poor' detected force remains where it is.

Design note: This represents intentional disinformation by NATO, fishing fleets, merchants transiting from one port to another, unimportant targets and so on.

30.7 Charlie II Class submarine (SS-N-9 'Siren' Missile) ^

Very slightly favors NATO, but 'historical'.

Charlie II class submarines can attack at short range only. When rolling for a missile attack, treat a long-range result as 'No attack'.

Design note: Not only was the Siren a slow, high-flying missile but it also had a range of only about 50 nautical miles. Expect success only against very lightly defended targets.

30.8 Fast Moving Submarines - noisy ^

Favors NATO slightly.

Any submarine which moves three zones during a movement action is marked with a 'Noise' marker and counts as one noise value worse (10 to 9, 9 to 8). If they have a save, they still retain it. They are unmarked when they become unSpent (at the end of the turn).

This doesn't apply to **Akula, Oscar, Alfa and Sierra** class Soviet submarines as they can dive extremely deep which would help to mask the noise generated.

30.9 Submarines operating alone ^

Favors Soviets slightly.

Submarines cannot use the Hydrophone rule to spend themselves and roll an extra die, instead always taking the option to roll a single dice and remain 'unSpent'.

Design note: This represents hunter-killers being difficult to contact and accustomed to operating alone.

30.10 Soviet Air to Air Refueling Capability

Favors NATO significantly, somewhat reduces the fun factor – but it was requested!

The SALT negotiations should have limited the number of heavy Soviet bombers. If you think the agreement was kept and the Backfires were eventually included in the heavy bomber designation stipulated by the talks, (and therefore had any refueling capability removed) you can use this rule. Soviet Tu-22 Backfires may not use the Tanker unit rules.

30.11 Soviet Planning

Favors Soviets.

If you want to play with some of these optional rules but feel that they lean a little towards NATO or else feel that the Soviet player needs a little bit of help in any case, use this option.

At the start of the game, the Soviet player may, one at a time, discard a card from his hand and draw another. Continue up to five times or until the Soviet player is happy to proceed.

30.12 Fighter Sweeps ^

Favors neither

An air Strike may be launched without any Strike component and may consist of only fighters. It is resolved in the same way – interceptors must intercept it and CAP must fight. At the end of fighter combat, the attacking fighters return to base 'Spent'. To fly a fighter sweep costs a single Operations Point regardless of the number of fighter units flying.

30.13 Shipwreck! ^

Favors Soviets slightly

When resolving Soviet red Missile hit selection roll an extra die to determine if the attacker can allocate the first step loss.

Design note Shipwreck apparently had advanced target detection and classification systems.

31.0 SURFACE SHIP UNIT REFERENCE

Because of the scale of the game, it was necessary to group ship types into formations. This table shows the component parts of each ship unit in the game. Each represents three units of its class, except where noted in brackets.

The units listed below as 'bd' can breakdown into the units marked 'breakdown'.

NATO Unit			
US-1	Ticonderoga Mk26 (1)	C.F. Adams	Knox
US-2	Leahy	Belknap	Knox
US-3	Virginia	Coontz	Knox
US-4 (bd)	Leahy	C.F. Adams	Knox
US-5 (bd)	Belknap	O.H. Perry	O.H. Perry
US-6 (bd)	California	O.H. Perry	O.H. Perry
US-7	Knox	Knox	
US-8	Spruance	Spruance	Spruance ½
US-9	Spruance	Spruance	Spruance ½
US-10	O.H. Perry	O.H. Perry	
US-11	Ticonderoga Mk26 (1)	O.H. Perry	O.H. Perry
US-12 (opt)	Ticonderoga Mk26 (1)	O.H. Perry	O.H. Perry
US-13 (breakdown)	Belknap	O.H. Perry	
US-14 (breakdown)	California	O.H. Perry	
US-15 (breakdown)	½ Leahy (1)	O.H. Perry	

US-16 (breakdown)	½ Leahy (2)	O.H. Perry	
US-17 (breakdown)	Knox	C.F. Adams	
German	Bremen	Lutjens	Weilingen (Bel)
Dutch	Tromp	Kortenaer	Kortenaer
Canadian	Restigouche	Iroquois	Knox (US)
Spanish	Descubierta	Descubierta	Baleares (knox)
Italian	Impavido/Audace	Maestrale	Lupo
French	Mixed Classes	Georges Leg.	Cmdt. Reviere
UK ASW	Leander (TA)	Leander (B IIIA)	Type 22
UK AAW	Type 42	Type 42	Type 22
UK ASW-2	Leander (TA)	Type 42 (B III)	Type 22 (B III)

Soviet Unit			
Sov-Udaloy	Sovremenny	Udaloy	Krivak I
Sov-Udaloy 1985+	Sovremenny	Udaloy Improved	
Krivak ASW	Krivak I x 2 (6)	½ Krivak II (2)	½ Mod Kildin (2)
Kara Group	Kara x 2 (6)	Kashin	Mod Kashin
Kresta Group	Kresta II	Kresta II	Mod Kashin
Kynda Group	Kynda	Kresta I	Krivak II, Kashin
Slava Group	Slava (1)	Krivak II	Kanin
Imp Slava	+1 Slava (1)		

32.0 ODDS CALCULATOR & MATH HELPERS

The tables below show the odds of rolling the indicated numbers by # dice being rolled.

2d10		7+	8+	9+	10
0		36	49	64	81
1		48	42	32	18
2		16	9	4	1

3d10		7+	8+	9+	10
0		22	34	51.2	72.9
1		43	44	38	24
2		29	19	10	3
3		6	3	0.8	0.1

4d10		7+	8+	9+	10
0		14	24	40.84	65.79
1		35	41	41	29
2		35	26	15	4.8
3		14	8	3	0.4
4		2	1	0.16	0.01

5d10		7+	8+	9+	10
0		7	16.93	33.33	58.85
1		26	36	41	33
2		35	31	20	7.3
3		23	13	5	0.81
4		8	2.83	0.64	0.04
5		1	0.24	0.03	0



Soviet Udaloy class destoryer

Blue Water Navy Campaign Game
Order of Battle 1.0
Soviet ORBAT 1983

Air	Ships	Submarines
Libya MiG-23 (3)	Kiev	Typhoon SSBN
Libya MiG-25	Kirov	Delta I/II SSBN (3)
Cuba MiG-21 (2)	Leningrad Moskva	Delta III SSBN (2)
Cuba MiG-23/29	Sverdlov	Yankee SSBN (2)
MiG-29 'Fulcrum'	Minsk	
Tu-95 Bear-G	Sov-Udaloy	Charlie II
Tu-16 Badger (6)	Krivak ASW (2)	Oscar I
Tu-22M Backfire (3) (1 in dead box)	Kara	Echo II (black) (2)
Tu-22M Backfire Reserve (3)	Kresta	Echo II (white) (3)
Tu-16N Tanker (2)	Kynda	Juliett (2)
Tu-95 Bear-D (3)	Slava 1983-85	Tango (TA)
Tu-95 Badger-D (2)		Tango
IL-38 (range 2)		Whiskey (3)
IL-38 (range 1)		Foxtrot (7)
Tu-95 Bear-F (3)	Syria SAM	Kilo
MiG -21 Black Sea	Cuba SAM	November (2)
MiG -23 Black Sea	Libya SAM	Alfa (2)
MiG -25 Black Sea	Baltic SAM	Victor I (4)
MiG-25 Baltic (2)	Kola SAM	Victor II (2)
MiG-23 Baltic (2)	Black SAM	Victor III (4)
MP Helos Baltic	Lourdes	Yankee SSN (4)
MP Helos Kola		Yankee SSN (TA)
MiG-25 Kola		Cuba Foxtrot
MiG-23 Kola (3)		Libya Foxtrot
MiG-31 Kola		
MiG-21 Syria		
MiG-23 Syria (2)		
Yak-38 Kiev		
Yak-38 Minsk		
RORSAT (9)		

1985 Additions to 1983 ORBAT

Air	Ships	Submarines
Tu-22M Backfire	Sov-Udaloy	Kilo
Tu-95 Bear-G		Yankee Notch
MIG 31 Kola		

1989 Additions to 1983 ORBAT

Air	Ships	Submarines
Kuznetsov Carrier Air Group	Imp Slava (replaces 1983-85 Slava)	Delta IV SSBN
Yak-38 Baku	Kalinin	Sierra
	Kuznetsov Special	Akula
	Baku	Victor III
		Kilo
		Oscar II
		Victor III (I)

NATO ORBAT 1983

Air	Ships	Submarines
UK Nimrod (range 1)	Saratoga	UK Oberon (2)
UK Nimrod (range 2)	Enterprise	UK Swiftsure
UK Buccaneer	Nimitz	UK Churchill
UK Phantom (2)	America	France Agosta (1983-85)
UK Sea Harrier (Invincible/Illustrious)	Kennedy	France Daphne
UK Phantom (1983 only)	Kitty Hawk	Italy Sauro
France Atlantique (range 1)	Iowa/New Jersey	Italy Toti
France Atlantique (range 2)	UK Invincible/Illustrious	Spain Daphne
France F8 (Foch)	France Foch	Germany Type 206 (2)
France Super Etend (Foch)	Italy V. Veneto A. Doria	Germany Type 205
France Mirage (2)	Spain Dedalo	Norway Kobben (2)
Spain Sea Harrier (Dedalo)		Netherlands Zwaardvis (1983-85)
Spain/Port P3	Spanish	Netherlands Dolfijn
Canada P3	Germany	Denmark Delfinen
Germany Atlantique	Dutch	Canada Oberon
Germany Tornado Strike (2)	Canada	Portugal Albacora
Germany Phantom (2)	Italy	Turkish Balao
Italy Atlantique	France	Turkish Type 209
Italy Starfighter (range 0)	UK ASW	Los Angeles (4)
Italy Starfighter (range 1)	UK AAW	Sturgeon (7)
Netherlands P3	US 1	Thresher (1983-85)
Norway P3	US 2	Ethan Allen (1983-85)
Bermuda P3	US 3	CAPTOR Mines (2)
USA F15 (2)	US 4	Breakdown Los Angeles (2)
USA P3 (2)	US 5	Breakdown Sturgeon (2)
Cuba Strike F15 (2)	US 6	
Cuba Strike F4 (2)	US 7	
Lajes P3	US 8	
F15 Eagles (2)	US 9	
Roosevelt Roads P3		
Keflavik P3	Breakdown US 13-17	
Keflavik 2		
Keflavik 1		
F14 Tomcat (6) (US carriers)		
Carrier Air Group (6) (US carriers)		

1985 Additions to 1983 ORBAT

Air	Ships	Submarines
UK Tornado (Replaces 1983 only Phantom)	US 10	UK Trafalgar
	US 11	Spain Agosta
	UK ASW-2	Los Angeles

1989 Additions to 1983 ORBAT

Air	Ships	Submarines
	Ticonderoga VLS (4)	France Rubis
		Los Angeles Imp
		Los Angeles VLS (2)
		Netherlands Zwaardvis (Replaces)
		France Agosta (Replaces)
		Remove US Thresher & Ethan Allen subs

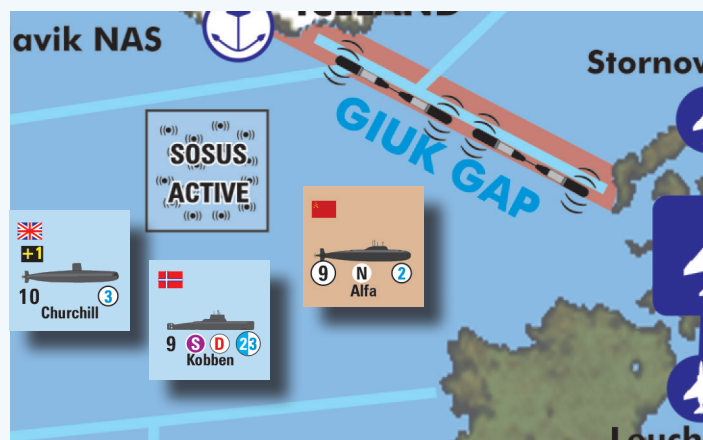


Netherlands Zwaardvis

EXAMPLES OF PLAY

Anti-Submarine Warfare #1

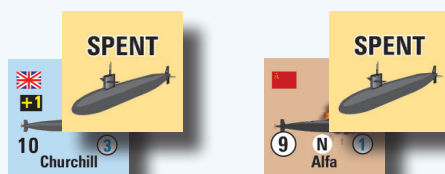
An Alfa class submarine performs a 'move and attack' action and moves two zones, crossing the SOSUS line. Per 6.8, NATO can interrupt the move and either roll a single die or roll a unit's normal amount of dice +1.



NATO wants to kill the Alfa, so decides to roll 4 dice with Churchill. The rolls are 1-5-8-9. Adding Churchill's Tactical Value of +1 to each roll, two hits are scored (8 becomes 9 and 9 is already a hit – Alfa has a defence of 9). Normally this would cause two step losses, killing Alfa, but she has the save notation which allows a 6+ roll against each hit to cancel it (*representing her great speed and deep diving capability*). The Soviet player rolls two dice – 5 and 7, cancelling one hit. Alfa is flipped.



Now, Alfa can attack. The reduced side of the counter has an ASW value of 1, so one die is rolled. Alfa rolls an 8, which has no effect. Alfa and Churchill are both Spent.



Anti-Submarine Warfare #2

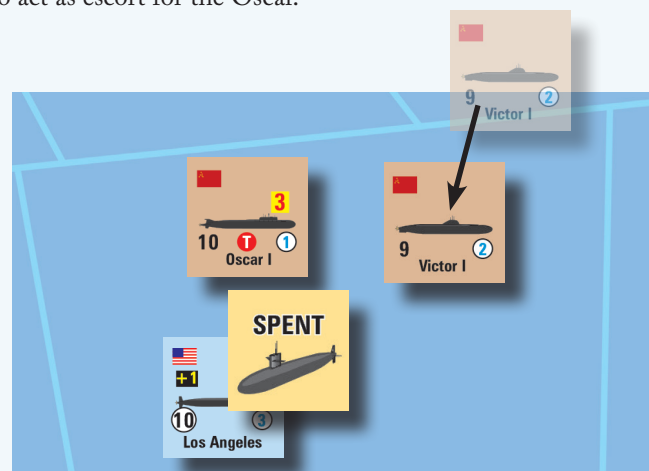
A Soviet Oscar class submarine is alone in a Sea Zone. NATO uses a 'move and attack' action with a Los Angeles submarine and rolls to attack.



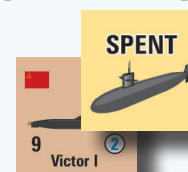
The roll is 5-5-8. With +1 Tactical Value this is not enough to kill Oscar (a 10 is required) and since NATO does not have rocket torpedoes, the double 5 has no effect. The LA is marked Spent.



In the Soviet activation, a Victor I is moved into the same zone to act as escort for the Oscar.



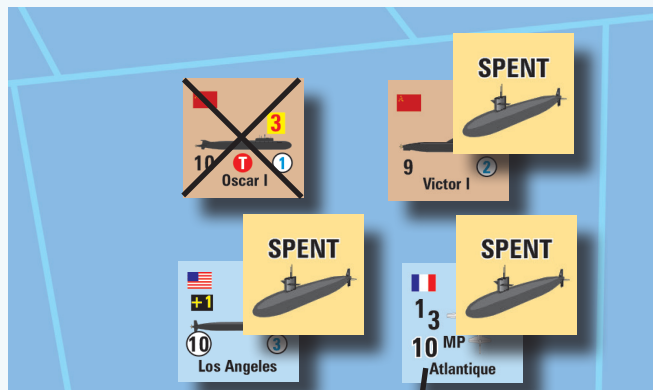
Victor attacks, rolling 6-6, which is a rocket torpedo attack. The only possible target is the LA, who has a save and rolls an 8, avoiding the rocket torpedo. Victor is Spent.



NATO uses a 'MP fly and attack' action and moves the Atlantique out to attack.



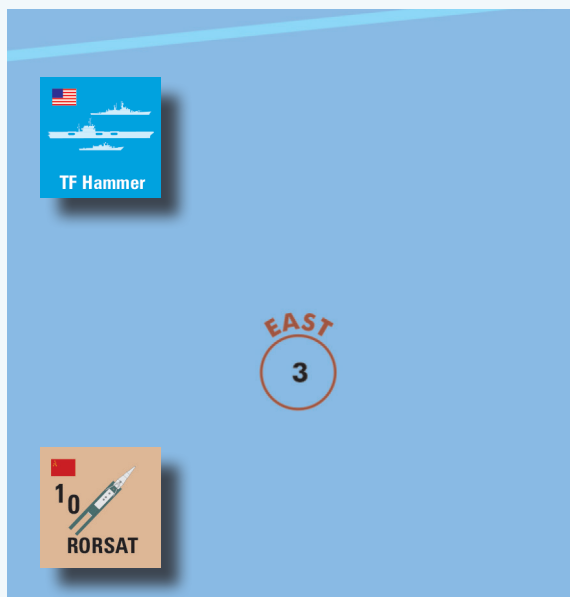
Atlantique rolls 1-8-10. The 10 is enough to flip either boat. Usually, the defender allocates the first hit but a natural 10 overrides this – so Atlantique can kill Oscar. She has no save and only a single step, so it is destroyed. After rolling Atlantique is marked Spent and returns to base.



Atlantique returns to base.

Detection of a Task Force

Task Force Hammer moves into a zone containing a RORSAT during movement.



The RORSAT rolls one die on the Task Forces Detection Table, rolling a 4 which is "Poor detection. May upgrade any Poor to Good. Placed by TF owner unless upgrading".

No detection can be upgraded (from Poor to Good) since there are no Detection markers. Therefore, the TF owner can place a Poor detection. If there were more than one Task Force in the zone, there would be a choice as to which to give away. Since there is not, TF Hammer is marked 'Poor' detection. A Poor detection reveals the presence of a Convoy and an F-8 Fighter. With this information, the Soviet knows that there is a French carrier but no US carriers (due to RADAR intercepts).

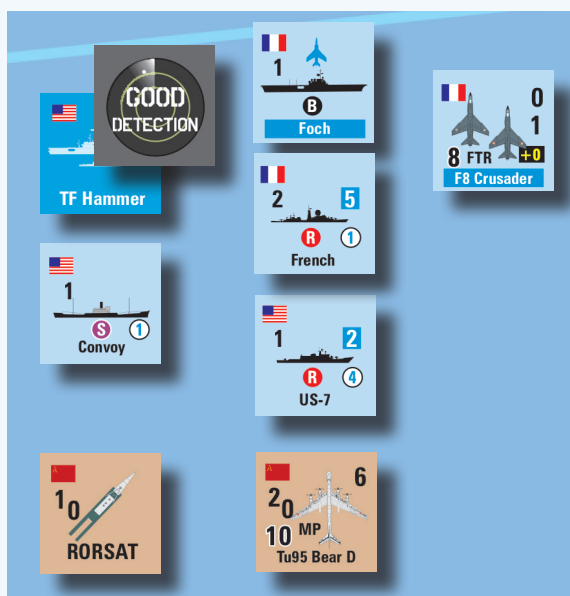


Maritime Patrol detection and CAP Response

The Soviet knows that a Good detection gives positive modifiers to all attack types and so in the Soviet player's next action, a Bear-D is flown to the zone. The F-8 may perform CAP prior to the Bear resolving its mission.



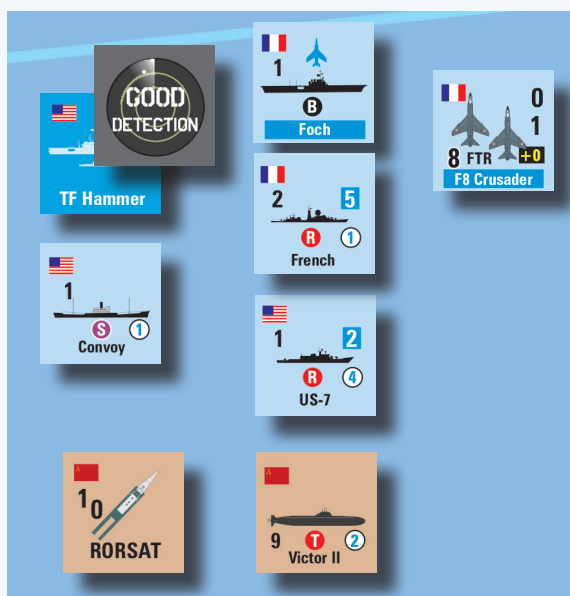
The F-8 will use 'Fighter vs Maritime Patrol' combat to try to shoot down the Bear. It rolls its number air to air dice (1) minus 2 dice, but to a minimum of one. So, one die. To this, it will add its Tactical Value of +0, so an unmodified die. The die is a 9, which causes the Bear to lose one of its two dice when searching – the Bears are put off by the F-8's constant harassment. Following this, the Bear rolls a 3 (with only one die instead of two) on the Task Forces Detection Table. The Poor detection can be upgraded to Good. This causes the Task Force make-up to be revealed. The Bear returns to its base and is marked Spent.



Tu-95 Bear-D moves away.

Normally, it would now be NATO's turn to play an Operations Point, but the MP 'fly and detect' action says "If a detection is placed or upgraded, active player may then play one further action to attack a detected TF" – so the Soviet spends one more Operations Point right away to move and attack with a Victor II class submarine.

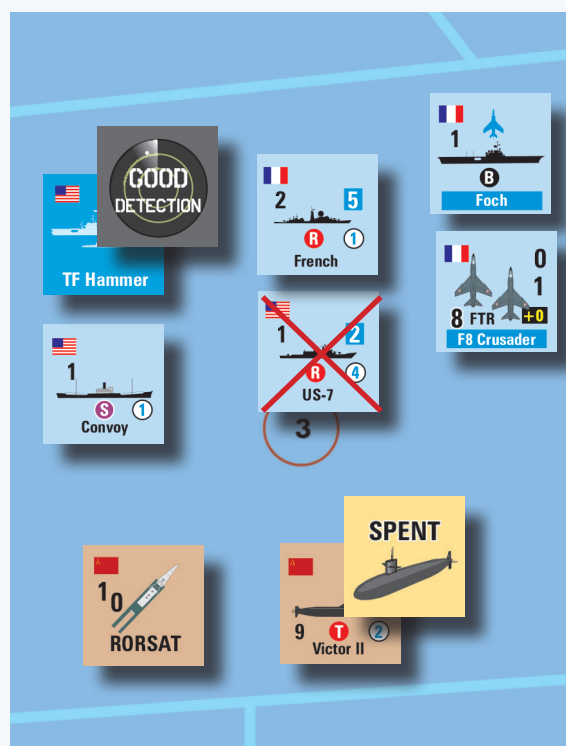
Submarine attack against a Task Force



The submarine will roll its two dice on the 'Submarine vs Task Force Attack Table, adding one extra die for the Good detection. The rolls are 5-6-8. The 5 is a miss and is discarded. The 6 is a hit but can be upgraded because of the (T) super heavy torpedoes carried by Victor to 'PM / Amph / Con'. And the 8 is the same result. This will allow two hits to the Convoy or any other non-capital ship in the Task Force.

Before hits are allocated the Task Force can defend itself – we add the ASW value of all the ships present 1-1-4=6 and then add +2 for the carrier's helicopters for a total of 8. This generates three dice each with no DRM.

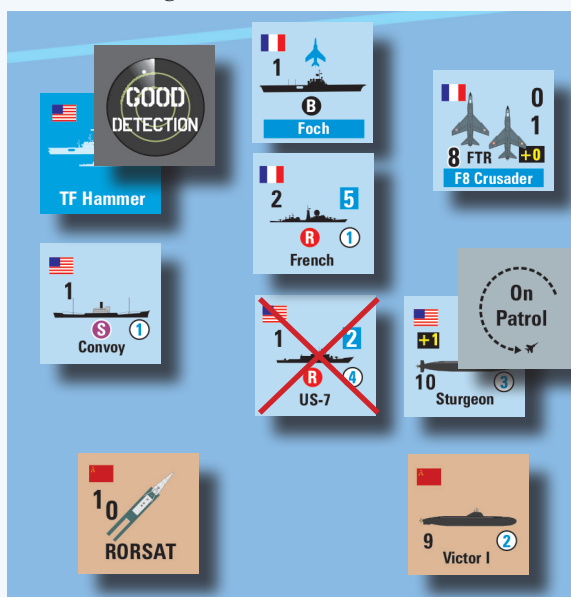
The dice are rolled and the result is 2-7-8. no die is high enough to kill the Victor, but any roll of 8 or above is a 'disrupt' which allows the TF to force to choose and cancel one attack die. The TF removes one of the dice that will produce a hit – so the TF succeeds in scaring away a Victor and disrupting its attack. The remaining hit can be allocated by the Soviet player who chooses to take out the excellent US-7 ASW unit which has only a single step. This will allow further submarines to face less ASW opposition. The Victor is marked Spent.



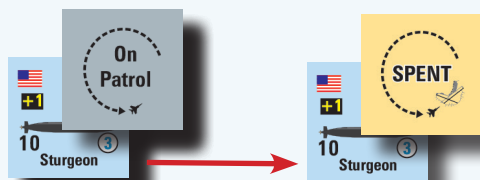
Victor II is scared away.

On-Patrol

It is now NATO's action. Keen to prevent a follow-up attack, a US Sturgeon is activated with a 'move and go on patrol' action. The submarine is nuclear and moves two zones, then is placed on patrol which will allow an interruption when any Soviet submarine moves into the zone. On the following activation, a Soviet Victor I is activated and moves to the zone with the intention of attacking.



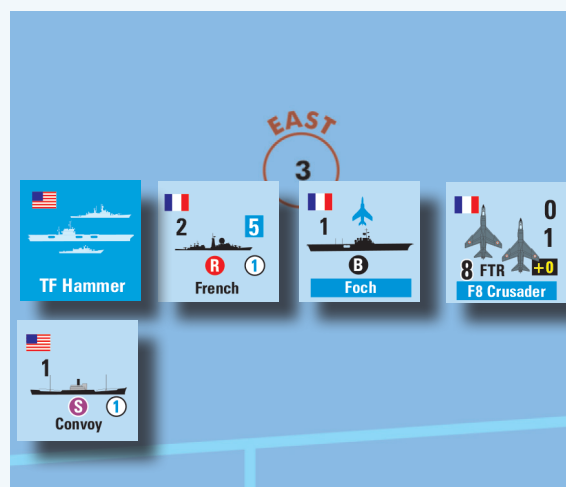
The Sturgeon can choose to interrupt and does so, flipping its On Patrol marker to Spent. It rolls three dice on the ASW Table: 5-6-8. With the +1, a 9 causes a step loss to Victor who now completes the attack with one less die. (the flip side of the Victor has only one die).



Missile attack by submarines

Later in the turn, an Echo II performs a 'move and attack' against the same TF and chooses to use missiles. Looking at the 'Submarine Launched Missiles vs Task Force' Table, the Echo rolls two dice because the TF is marked Good detection. The highest die is a 9 which is a short-ranged attack. This will give the Echo's missiles a better chance of reaching their targets. The Echo is marked Spent and Cruise Missiles Fired and then we resolve the effect.

Missile Attacks Against Task Forces



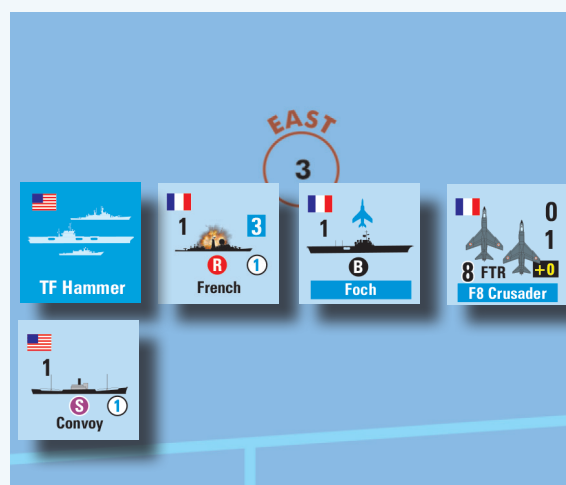
The NATO Task Force has a total of 4 SAM points, with three incoming missiles the table shows we can only fire +1 SAM more than there are missiles incoming. Since there are three incomings, we can fire four SAMs

(This is a representation of ammo, mounts bearing, multiple missiles being fired at incoming missiles and so on). However, all SAM shots are penalized by a DRM of -2 due to the short-ranged attack.

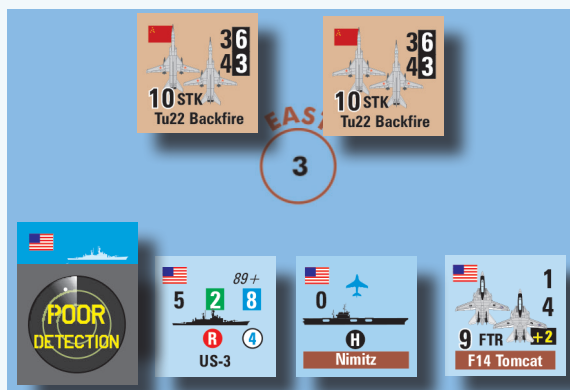
The NATO player rolls 4 dice: 1-4-5-8. Only 2 missiles are downed, and so 1 remains – the Soviet player rolls a single die, scoring a 5, +1 for Good detection is 6. The missile hits something.

To determine what is hit, the Soviet player rolls two dice needing a 9 or 10 on either die. The roll is 3-5, and so NATO allocates the hit – choosing to flip the French fleet and save the Convoy.

Had the Soviet roll included a 9 or 10, the Convoy could have been chosen – and missiles cause two hits each to a Convoy, or else the French carrier Foch could have been chosen.

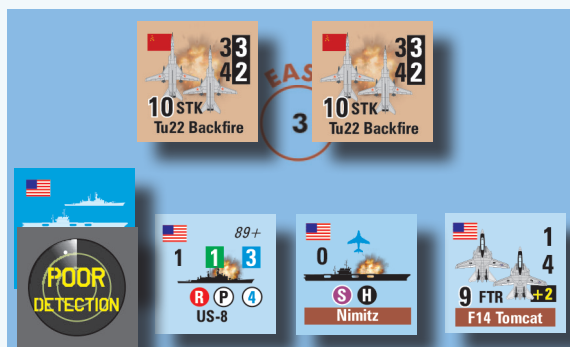


Two Regiment Missile Attack on a Task Force



Two Backfire regiments are attacking a US Task Force at range 3. The F-14's roll 1-6-8-10, with a +2 modifier for their Tactical Value. That is two hits scored, since $8+2 = 10$ and one natural 10 result. If the Task Force been marked Good detection, the strike air would receive a 6+ save against each hit, but because they are not there is no saving throw – For the 10 result, the Soviet flips a Backfire and that step does not attack. Since the flip side of a two-step backfire is 3 missiles at range three the counter only attacks with 3 missiles.

For the 8+2, the Soviet player flips the other Backfire (or could remove the flipped one) but as it's not a natural 10 this takes place after the missiles are launched.



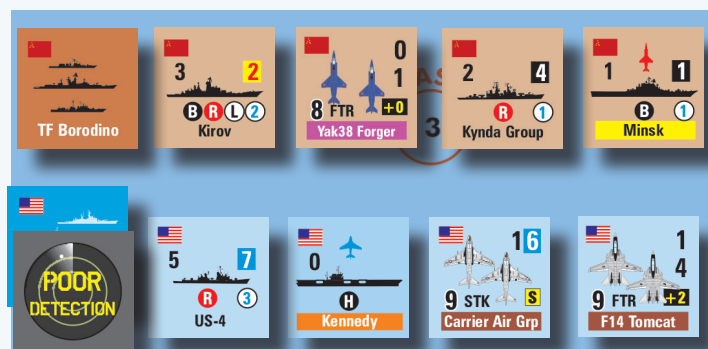
That's 9 missiles in the air. Additionally, the roll of a 1 with F-14's shoots down one incoming missile, so there are now 8 missiles. This can be tracked as you calculate with the appropriate Missiles marker.

NATO fires 5 SAMs, shooting down 5 missiles. The Soviet player rolls a 1 for the remaining pair and a 5 for the single remaining missile, for a total of two hits scored. The Soviet player rolls two dice for allocation and manages a 9 which allows the first missile to be allocated by the Soviet player, who picks the carrier. The second missile, NATO allocates to US-8, which is flipped.

Consulting the capital ship damage rules, 2 dice are rolled for a Soviet black missile – a 5 and an 8. A US carrier is (H) Huge and requires a 10 to be sunk, so the carrier is damaged and flipped. There are 4 steps of Air Units on the carrier currently – two F-14 and two CAG, so half of them (two) must roll and take a step loss on 6+.

Task Force vs Task Force

During a ships move event, these two Task Forces may both move. The NATO force is marked Poor detection and so moves first. The Soviet force then uses its move to move into the same Sea Zone.



All detections are then degraded, so both Task Forces are now undetected.

Then fighter vs fighter combat due to the movement of Task Forces is resolved [17.6] – there is a Soviet Yak-38 and a US F-14 in the same zone together so they will fight.

Looking at the odds, the Soviet player would receive one die with a +0 versus 4 dice at +2. The Yaks are horrendously outmatched. As a result, the Soviet decides to avoid combat and places a Spent marker on his Yaks - they refuse to meet the F-14's and will not be able to counter the US intrinsic detection capability.



Next, intrinsic detection units are attacked and then they will search.

The US has one carrier, and the Soviet player has one carrier. Only aircraft carriers have intrinsic detection assets [14.3] and the Soviet player cannot attack the US asset. The US can attack the Soviet asset as NATO still has active fighters in the zone. The F14's use the Fighter vs MP table, rolling two less dice than normal ($4-2=2$) scoring 4-7 with a +2 added to each die. This is enough to cause the Soviet detection asset to lose its only die (6+ is required vs a helo to cause a die loss). The US carrier asset rolls two dice (US carriers always roll two dice) 1-4 for detection. The one has no effect but the 4 places a Poor Detection marker on the Soviet TF.



The move event is now over, and we return to the next action which is a Soviet OP, hence the aggressive move. The Soviet player is currently active and so can use a Soviet spy. The player places a Poor Detection marker on the US Task Force and rolls

for spy survival. Following this the Soviet has a few options – to use a maritime detection asset to try to place a Good detection, or a submarine – but subs need natural tens. Or, take a chance and try for a missile shot. The Soviet ponders, downs a quick Vodka and reaches for the die.



A 6+ is required to launch a TF missile attack with yellow and black missiles against a poor detected TF, and this is only allowed because the Soviet force is fast [17.14] – the Soviet rolls the die and scores a ten. The missiles fly!

There are 2 yellow and 5 black missiles, which are marked on the Missiles Track with the appropriate markers. The firing ships are marked as being out of ammo. Because an F-14 is present the NATO player can roll one die (as the detection status of the Soviet TF is poor) a roll of a one shoots down a single missile – red is selected.

See [17.18] for multiple missile types attacking together but essentially, the defender can choose what to shoot at, but does so before rolling each die. NATO has 5 SAMs and can use them all since the number of defending SAMs is not more than the number of incoming missiles.

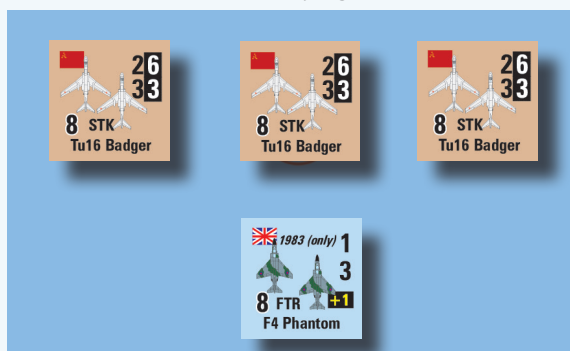
The first 4 SAMs, NATO rolls against black missiles – 1,3,4,9 for a total of 4 hits. There are now 1 yellow and 1 black missile remaining. Since there are easier targets left, (black is easier to hit) NATO rolls another die at them, again rolling a 1. No hits. The Soviet player now rolls for effect. Since two missiles remain, one die is rolled, a five causes a single hit.

In the NATO action, assuming the carrier is afloat, NATO could launch an air strike or call in other assets to attack the Soviets. They can only use their own ship-borne missiles via card play or in the Baltic Sea as the range is otherwise assumed to be too far for *Harpoon*. See 17.14. This section explains the working of ship launched missiles.

Missiles vs a Land Target

The Soviets launch a strike with 3 Tu-16 Badger regiments targeting the UK airfields.

The British have one Phantom on interception in North 5-6 and two at the airfield and therefore flying CAP.



The Soviets trace the route to the target zone and find it clear of enemy aircraft until they arrive in North 5-6. Since the Soviet units entered a zone with a unit on an Interception mission, interception is triggered. One fighter unit can roll per strike unit present, but since there is only one UK fighter, only it can roll – if there were 3 UK fighters, all three could roll – one per strike unit.

The Phantom will roll 3 die and add +1 to each roll, but because it is intercepting it loses one die. A modified total of 8 or higher will shoot down a step of Badgers.

The rolls are: 2-10. One Badger is flipped and because this happens during Interception there is no saving throw, and it happens before the Badgers perform their attack – so less missiles will now fly.

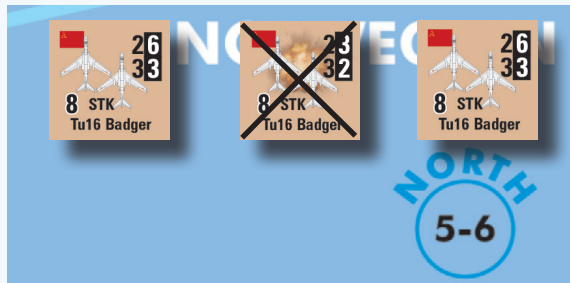


Lastly, because the strike has been detected by interceptors (any shoot down or failure to shoot down by one pip on the die by interceptors is a detection) the CAP will be more effective – they are aware of the threat.

The Soviet player can immediately abort the raid if desired after all Interception has resolved. For the purposes of the example, we'll continue but in the real game an abort once detected is sometimes a wise choice.

Next, we resolve CAP. There are two Phantoms available, and with at least two inbound strike units both can roll. Each rolls 3 dice, up to 4 dice for the detection, with a +1 Tactical Modifier.

A total of three 8+ results are rolled. Because this is CAP, the Soviet receives a saving throw. CAP from an airfield grants a 6+ save. The Soviet rolls three dice and saves two, so one more Badger step is destroyed. However, because the attacking roll was not a natural ten the shot down step launches missiles before being killed.



At the end of air to air combat, 15 Soviet missiles are in the air. Consulting the Soviet Missiles Table, 13 or more missiles grant three dice on the Cruise Missile Attack Table. 1-6-7 is rolled. The 1 has no effect, the 6 is 1 hit and the 7 is 1 hit and 1 kill. Two hits are placed on the UK airfields, which places a Light Damage marker because the number of hits is equal or greater than one but not equal to three (the UK facility damage values). A Light damage on an airfield had the following effect: “No air may fly, return to base on patrol interceptors and MP units and mark Spent. Also applies a CAP modifier” The interceptor is returned to base and marked Spent. A Nimrod on patrol some distance from the UK is likewise returned and marked Spent.

Lastly, one Air Unit step is destroyed on the ground. Odd kills are allocated by the attacker and so the Soviet player flips a unit of Phantoms and then returns the remaining Badgers to their base.



Bombs vs Soviet Troops

A US Task Force is operating in the Mediterranean and wants to bomb the Southern European Soviet armies. They are operating in Med 3-4, which is one zone away from Med 5-6, adjacent to the Troops icon. Since US CAGs have a range of one, this is in range. The NATO player pays two OPs to fly 2 x US CAGs. One F-14 will escort them, leaving one over the fleet. The escort is free to fly.

The striking Soviet Troops process is followed – the F-14 rolls 4 air to air dice adding +2 to each roll. Three rolls of 8+ are counted which gives a result of “1 attack vs each striking Air Unit”. There are two striking units, so two dice are rolled: 5-6. A 6+ ignores one of the US bombing dice. A nine or ten would have been required to cause a step loss to a CAG (the CAGs defence value). This process represents Soviet fighters and SAM systems over the front protecting the troops.

There are 4 steps of CAG which means 4 bombing dice, but one was cancelled so three dice are rolled on the aircraft bombing table: 3-6-9, for a total of 3 hit results. Three hits are placed on the troops space on the map and the strike and fighters return to base spent. Three hits are required to prevent an advance on the War Track, so the NATO player has succeeded in temporarily halting the Soviet forces.



DESIGNERS NOTES

This is the first game for a mass audience I've made and as such it's been a bit of a trial by fire – I started with an idea of what I wanted but not a clear scope, and as with all projects without a clear scope, I ended up doing several times the work! At one point we had individual submarines, but it was huge and unwieldy. There were battle boards, lengthy chrome and extensive detail – you name it, we had it. What we have now is a much stripped-down version of those initial attempts – I didn't want something that took 40 hours to play. The final game still has plenty of 'modern' naval feel, and a fair amount of chrome but can be played in 8-12 hours whilst still retaining the feel that you're fighting with submarines and aircraft carriers.

I made the game for the same reasons most people give – there wasn't a game that already did the job – I wanted a modern system with little down-time between each players chance to act, I wanted the whole hemisphere of the war rather than just a small part of it, with a realistic ORBAT using the details we now know, and I wanted it to be card driven because that reduces player workload – much of the chrome can go onto cards giving the same overall hi-tech chrome effect but without a million rules to account for everything. Finally, and most importantly, I wanted to be able to understand what everything means in the game – If I'm resolving 20 points of missiles, I want to know what that means in real terms – how many missiles is it, why is it 20 points instead of 15? – everything in Blue Water Navy calculates to a fixed value of something in the real world – the math might not always be spot on – it still has to do all of that at a high level and in 12 hours, but it's pretty close – the table at the end of these notes gives all of this information for those interested

I'll not go into the specifics of everything, but I will detail a few queries I've been asked over the course of development: submarines, why can they teleport 1000nmi (2 zones) and attack? –The scope of the game at 2 days per turn means that a nuclear submarine could make that distance at a cruising speed. The counter's location on the map should really be thought of as it's 'last known position' – for both sides. This allows a lot of fluid gameplay, but the 'On Patrol' mechanics also allow trap setting.

Why does the SOSUS line cover across the North Sea? This is really to account for the significant number of 'below the scale of the game' NATO units that would be present in the area.

Why are some of the NATO navies so small? The French had many more ships than that – indeed they did, but again we zoom out to the scale of the game and ignore everything that would have been on port sanitization, mine clearing, cross-channel escort, and so on. Plus, much of the not-represented equipment doesn't have a weapon on this scale – many small escorts still had ASW mortars, guns and chaff and nothing else. It's worth mentioning that for the same reason the several hundred Soviet fast attack craft are likewise 'zoomed out' but do make an appearance through cards and specific rules.

Why can't I fire all my SAMs at an attacking missile? Real navies must be ready for the next attack, and the next and so ammo

conservation is a part of it, but there's also an aspect of over-killing – a missile shot at by two SAMs and killed by the first wastes one round, which would have been better fired against a miss instead, and means the rail has to reload and so is out of action for a number of seconds. Plus coordinating the fire of 20 or more ships is a hugely complicated affair not even including the idea of where each ship is, closing or crossing shots and so on. Not everyone is in the right place to engage the threat.

Why are the Soviet submarines so poor compared to NATO submarines? Actually, I think they're probably a little over rated in places. The short answer is the Soviets were horribly outmatched in anti-submarine warfare in the early 1980's – NATO was quieter, had better detection, and had better weapons – the Soviets had some very exciting technology such as the Alfa submarines, but she was loud and easily tracked. The poorer earlier Soviet submarines maybe should have had only a single ASW die at full strength instead of two, but honestly that's not as much fun – and in any case NATO was never really sure how well the Soviets could hear, so the two value works for me.

Why are the Soviet diesel submarines so good? Well the 4-value ones represent 6 submarines instead of the usual 3 per counter, so they hit hard. The downside is those 6 still have only two steps so they die twice as fast as everyone else. Plus, they are really slow and it's a long way to the convoy lanes.

Why doesn't NATO get any bonus for using nuclear weapons at sea? NATO technology gives it the edge by default – by this period NATO was concentrating on conventional weapons – the days of 'nuclear everything' in the 60's and 70's having gone by the wayside – the theory being that if your weapon is sufficiently deadly anyway then it doesn't need to be nuclear in the first place. In any case nuclear torpedoes are inherently bad for submarines and those hunting them – they blind everyone due to the earth-shattering detonation.

As a final note, I am active on both Consimworld and BoardGameGeek in the Blue Water Navy folders – so if you have questions or comments, please come and find me there. There may well be free value-add scenarios, play videos etc. that appear so it's worth a check in any case. Thanks for playing and enjoy the game.

Stuart Tonge, Rotherham, England, February 2019

In Game	Real World	In Game	Real World
1 SAM	16 surface-to-air missiles	1 Torpedo Hit	~3-4 ships hit/damaged
1 missile	8 anti-ship or cruise missiles	1 Missile Hit	~3-4 warships hit/damaged or ~6-8 merchants hit/damaged
1 green cruise missile	~45 cruise missiles	1 ASW die (per die)	Coverage of about 30nmi of ocean with detection assets
1 fighter air to air die	~5 fighters performing an activity		



CREDITS

1	2	3	4	5	Small Scenario Ops Track
		Ships	Repair		
6	7	8	9	10	
	Ships	Repair	Fast		

Turn Sequence

- 1) Convoy placement (one in US & check for Med convoy)
- 2) Reinforcements
- 3) Operations
 - a) Soviet Card
 - b) NATO Card
 - c) Soviet first action
 - d) Igo-Ugo actions; Fast, Ships, Repair, Next Day may trigger
- 4) End of turn
 - a) On patrol units may return to base
RTB all air if both sides fighters occupy same zone
 - b) Remove all 'spent' & choose to remove 'on patrol'
 - c) Invasions
Resolve Denmark & Norway tracks - *2 advances possible*
Increase Danish mining if no amph landing
Danish strait de-mining & Norwegian surrender
 - d) War track changes / Soviet advances - *always 1 advance*
NATO nuclear weapon use
Capture facilities
Remove submarines.
Remove 3 hits on ALL tracks
 - e) If a first strike turn, +1 FSP if 6 payload in FS zones
 - f) Remove all weather & roll for new weather
 - g) NATO may hold 2, Soviet 1, then draw to 5 cards
 - h) Move the turn marker
 - i) Roll for Soviet ASAT point removal. 6+ removes 1 point